



Machine Learning 1

RECAP: Regression via MAP

Erik Bekkers

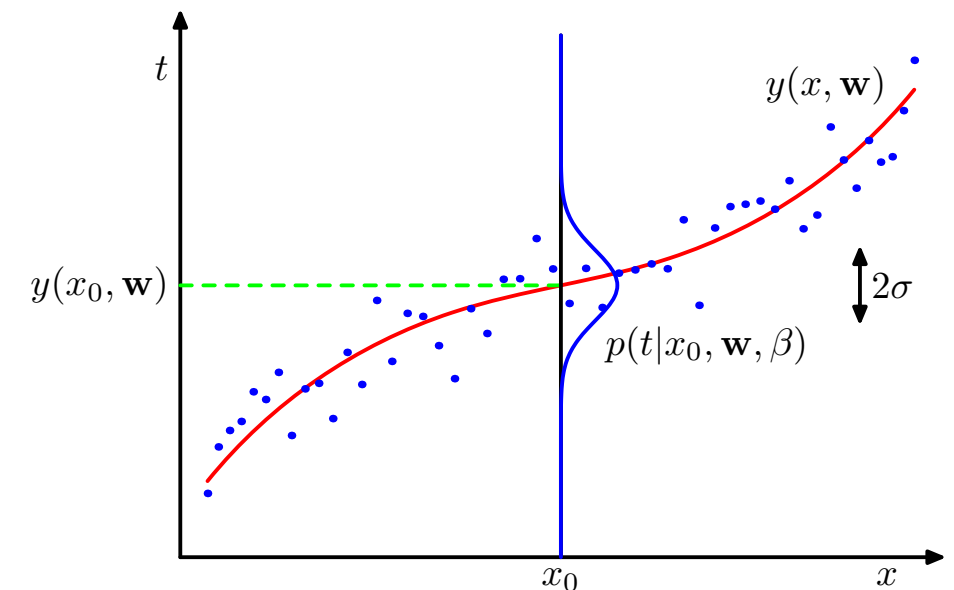
(Bishop 1.2.5 - 1.2.6)



MAP - Example Gaussian distribution

- ▶ **Likelihood**/Data model:

$$\begin{aligned} p(t | x, \mathbf{w}, \beta) &= \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1}) \\ &= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2} (t - y(x, \mathbf{w}))^2 \right] \end{aligned}$$



- ▶ **Prior** (assumption: $\mathbf{w}$ takes on small values with certainty α):

$$\begin{aligned} p(\mathbf{w} | \alpha) &= \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha^{-1}) \\ &= \left(\frac{\alpha}{2\pi} \right)^{M/2} \prod_{i=1}^M e^{-\frac{\alpha}{2} w_i w_i} \\ &= \left(\frac{\alpha}{2\pi} \right)^{M/2} e^{-\frac{\alpha}{2} \mathbf{w}^T \mathbf{w}} \end{aligned}$$

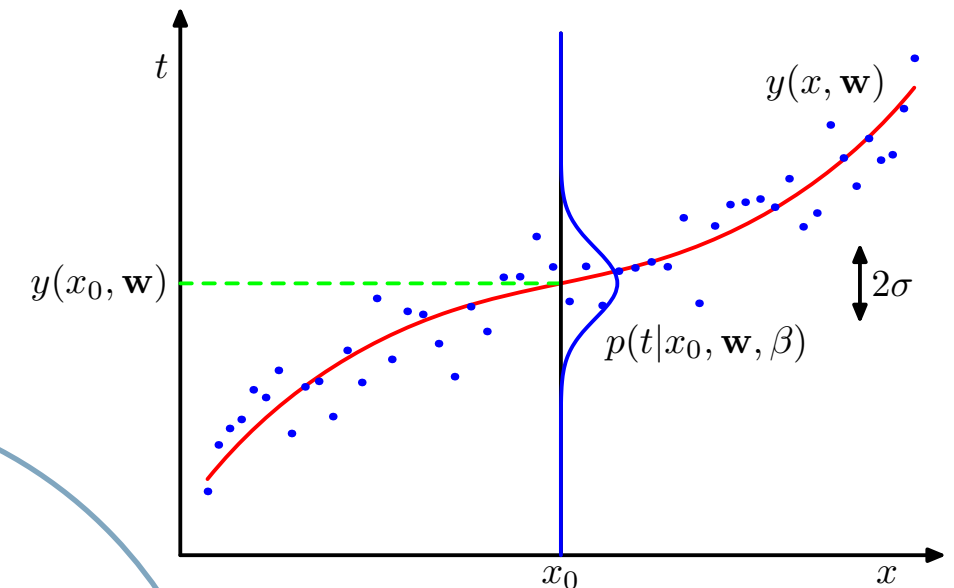
- ▶ **MAP solution** obtained by solving:

$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \quad -\log p(D | \mathbf{w}, \beta) - \log p(\mathbf{w}, \alpha)$$

MAP - Example Gaussian distribution

- ▶ **Likelihood**/Data model:

$$\begin{aligned} p(t | x, \mathbf{w}, \beta) &= \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1}) \\ &= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2} (t - y(x, \mathbf{w}))^2 \right] \end{aligned}$$



- ▶ **Prior** (assumption: $\mathbf{w}$ takes on small values with certainty α):

$$\begin{aligned} p(\mathbf{w} | \alpha) &= \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha^{-1}) \\ &= \left(\frac{\alpha}{2\pi} \right)^{M/2} \prod_{i=1}^M e^{-\frac{\alpha}{2} w_i w_i} \\ &= \left(\frac{\alpha}{2\pi} \right)^{M/2} e^{-\frac{\alpha}{2} \mathbf{w}^T \mathbf{w}} \end{aligned}$$

$$\begin{aligned} \log p(D | \mathbf{w}, \beta) &= -\frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 \\ &\quad + \frac{N}{2} \log \beta - \frac{N}{2} \log 2\pi \end{aligned}$$

- ▶ **MAP solution** obtained by solving:

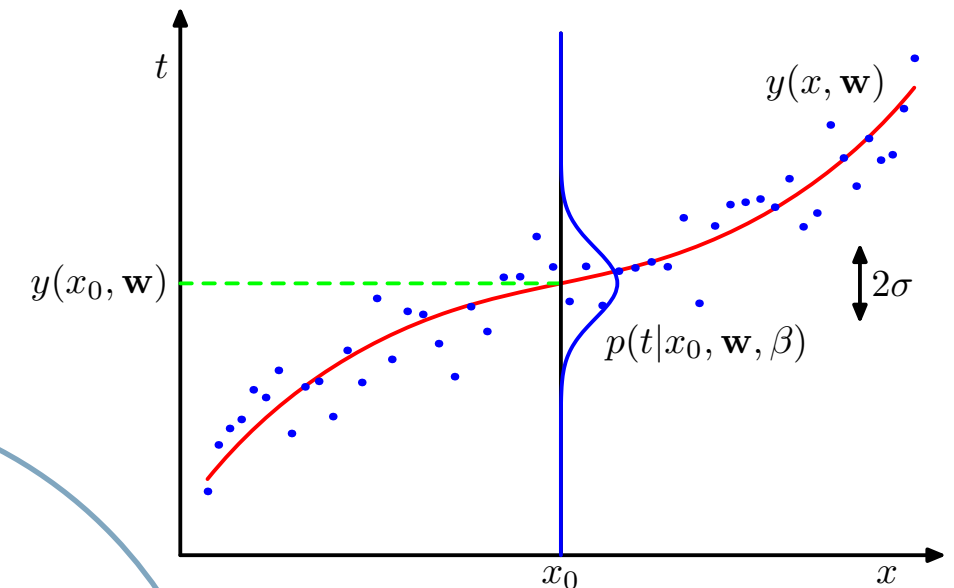
$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 - \log p(\mathbf{w}, \alpha)$$

MAP - Example Gaussian distribution

- ▶ **Likelihood**/Data model:

$$p(t|x, \mathbf{w}, \beta) = \mathcal{N}(t|y(x, \mathbf{w}), \beta^{-1})$$

$$= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2}(t - y(x, \mathbf{w}))^2 \right]$$



- ▶ **Prior** (assumption: $\mathbf{w}$ takes on small values with certainty α):

$$p(\mathbf{w}|\alpha) = \prod_{i=1}^M \mathcal{N}(w_i|0, \alpha^{-1})$$

$$= \left(\frac{\alpha}{2\pi}\right)^{M/2} \prod_{i=1}^M e^{-\frac{\alpha}{2}w_i w_i}$$

$$= \left(\frac{\alpha}{2\pi}\right)^{M/2} e^{-\frac{\alpha}{2}\mathbf{w}^T \mathbf{w}}$$

$$\log p(D|\mathbf{w}, \beta) = -\frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2$$

$$+ \frac{N}{2} \log \beta - \frac{N}{2} \log 2\pi$$

$$\log p(\mathbf{w}|\alpha) = -\frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

$$+ \frac{N}{2} \log \alpha - \frac{N}{2} \log 2\pi$$

- ▶ **MAP solution** obtained by solving:

$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

Parameter estimates with Gaussians

- ▶ Given **Likelihood**/Data model:

$$p(t | x, \mathbf{w}, \beta) = \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1})$$

- ▶ The **ML parameter estimate** is obtained via **least squares**:

$$\mathbf{w}_{ML} = \operatorname{argmin}_{\mathbf{w}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2$$

- ▶ Additionally, given Gaussian weight **Prior**:

$$p(\mathbf{w} | \alpha) = \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha^{-1})$$

- ▶ The **MAP parameter estimate** is obtained via **regularized least squares**:

$$\mathbf{w}_{MAP} = \operatorname{argmin}_{\mathbf{w}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$



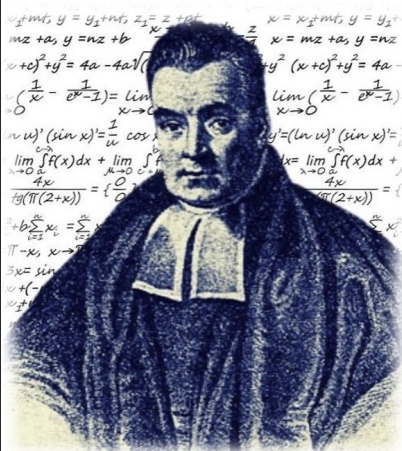
Machine Learning 1

Lecture 4 - Gaussian Posteriors - Sequential
Bayesian Learning - Bayesian Predictive
Distributions - Model Selection - Bias
Variance Decomposition



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The probabilistic view



W1

Probabilistic modeling

Random variables, distributions,
maximum likelihood estimation,
MAP

W7

Gaussian Processes

W6

Kernel Methods

W5

Latent variable models

W4

Neural Networks

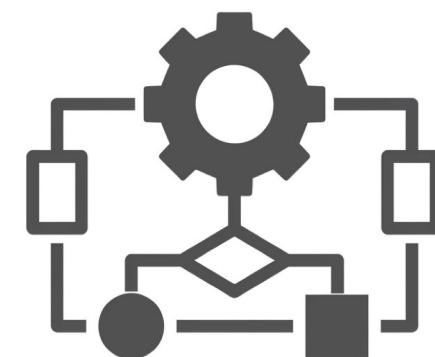
W3

Classification

W2

Regression

The algorithmic view



Optimization

Exact/analytic, (stochastic
gradient descent, constrained
optimization, 2nd order)

Mathematical tools

Multi-variate calculus, linear
algebra



Machine Learning 1

Lecture 4.3 - Supervised Learning
Bayesian Linear Regression - **Gaussian**
Posteriors

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(Bishop 3.3.1 (and 2.3.3))

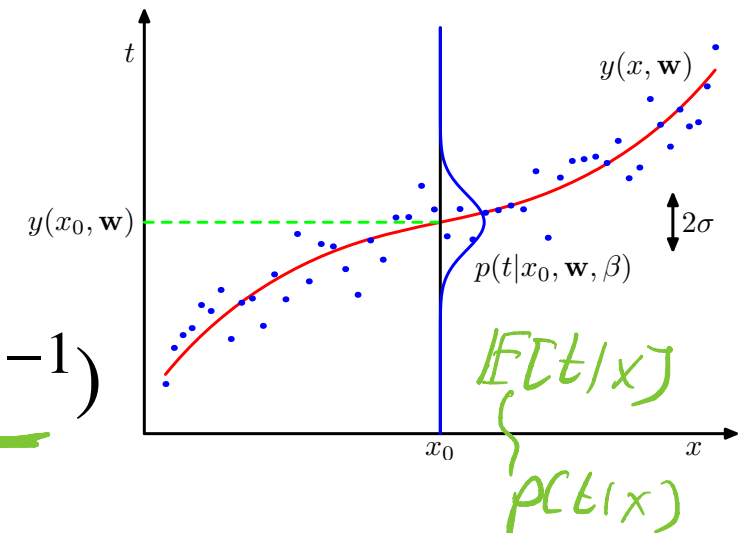


Bayesian Linear Regression

- Regression problem with:

- Data: $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T$, $\mathbf{t} = (t_1, \dots, t_N)^T$

- Predictive distribution $p(t' | \mathbf{x}, \mathbf{w}, \beta) = \mathcal{N}(t' | \underbrace{\mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}')}_{y(x')}, \underbrace{\beta^{-1}}_{\sigma^2})$



- Probabilistic model with Gaussians:

- Likelihood: $p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_n), \beta^{-1}) = \mathcal{N}(\mathbf{t} | \underbrace{\Phi \mathbf{w}}_{\boldsymbol{\mu}}, \underbrace{\beta^{-1} \mathbf{I}}_{\Sigma})$

- Conjugate prior: $p(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{m}_0, \mathbf{S}_0)$

- Posterior: $p(\mathbf{w} | \mathbf{t}, \mathbf{X}) = \frac{p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta)p(\mathbf{w})}{p(\mathbf{t} | \mathbf{X}, \beta)} = \mathcal{N}(\mathbf{w} | \mathbf{m}_N, \mathbf{S}_N)$

- Maximum A Posteriori estimate:

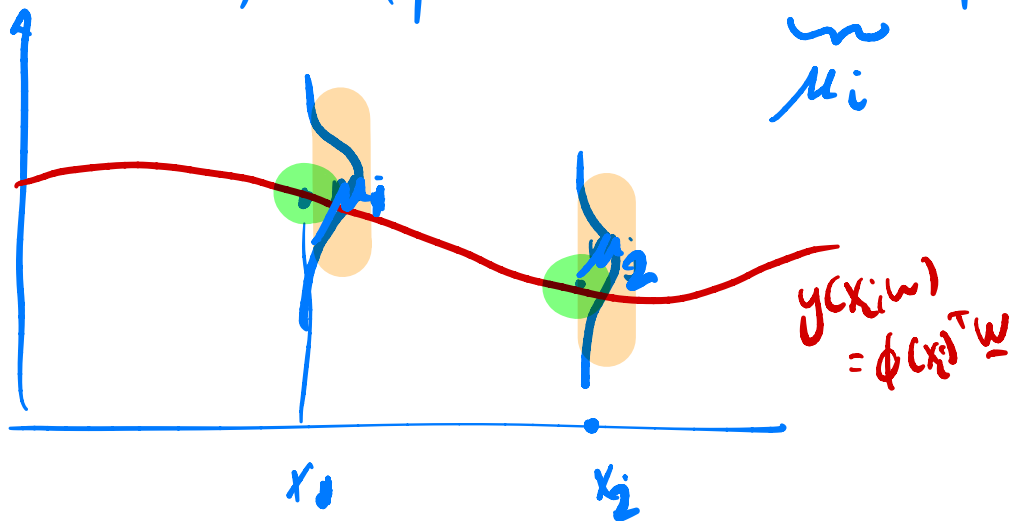
- $\mathbf{w}_{\text{MAP}} = \underset{\mathbf{w}}{\text{argmax}} p(\mathbf{w} | \mathbf{t}, \mathbf{X}) = \mathbf{m}_N$

Bishop Ch 2.3, Eq. 2.116

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \beta \Phi^T \Phi$$

$$\mathbf{m}_N = \mathbf{S}_N (\mathbf{S}_0^{-1} \mathbf{m}_0 + \beta \Phi^T \mathbf{t})$$

$$(*) \quad p(t_i | x_i, \underline{w}, \beta^{-1}) = \mathcal{N}(t_i | \underbrace{\phi_i^T \underline{w}}_{\mu_i}, \beta^{-1})$$



$$\underline{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} \phi(x_1)^T \underline{w} \\ \phi(x_2)^T \underline{w} \\ \vdots \end{pmatrix} = \begin{pmatrix} \phi_1(x_1) & \phi_2(x_1) & \dots \\ \vdots & \vdots & \ddots \\ \phi_1(x_N) & \phi_2(x_N) & \dots \end{pmatrix} \underline{w}$$

$N \times M$

$$= \Phi \underline{w}$$

predict all means
at once

$N \times N$

$$\Sigma = \beta^{-1} \mathbf{I}_N = \begin{pmatrix} \beta^{-1} & 0 & \dots \\ 0 & \beta^{-1} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

Since independent and shared β

$$(*) \prod_{i=1}^N e^{-\frac{\beta}{2} (t_i - \underline{\phi}_i^T \underline{w})^2}$$

$$= e^{-\frac{\beta}{2} \sum_{i=1}^N (t_i - \underline{\phi}_i^T \underline{w})^2}$$

$$= e^{-\frac{\beta}{2} (\underline{t} - \underline{\Phi} \underline{w})^T (\underline{t} - \underline{\Phi} \underline{w})}$$

$$= e^{-\frac{\beta}{2} (\underline{t} - \underline{\Phi} \underline{w})^T \underline{I} (\underline{t} - \underline{\Phi} \underline{w})}$$

$$\text{so } \underline{\mu} = \underline{\Phi} \underline{w} \quad \underline{w} \in \mathbb{R}^N$$

$$\Sigma = \beta^{-1} \underline{I}_N \quad \underline{I}_N \in \mathbb{R}^{N \times N}$$

Bayesian Linear Regression

- ▶ Special simple prior:

- ▶ $p(\mathbf{w} | \alpha) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I})$ ($\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{S}_0 = \alpha^{-1} \mathbf{I}$)

- ▶ Posterior

- ▶ $p(\mathbf{w} | \mathbf{X}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(\mathbf{w} | \mathbf{m}_N, \mathbf{S}_N)$

- ▶ With $\mathbf{m}_N = \mathbf{S}_N (\cancel{\mathbf{S}_0^{-1} \mathbf{m}_0} + \beta \Phi^T \mathbf{t}) = \beta \mathbf{S}_N \Phi^T \mathbf{t}$
 $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \beta \Phi^T \Phi = (\alpha \mathbf{I} + \beta \Phi^T \Phi)$

Bayesian Linear Regression

More-Penrose

- Limiting cases of the posterior

- $p(\mathbf{w} | \mathbf{X}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(\mathbf{w} | \mathbf{m}_N, \mathbf{S}_N)$ with $\mathbf{m}_N = \beta \mathbf{S}_N \Phi^T \mathbf{t}$
 $\mathbf{S}_N^{-1} = \alpha \mathbf{I} + \beta \Phi^T \Phi$

- Infinitely broad prior ($p(\mathbf{w} | \alpha) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I})$, $\alpha \rightarrow 0$):
 (No restriction/assumption on $\mathbf{w}$)

pseudo inverse

- $\lim_{\alpha \rightarrow 0} \mathbf{m}_N = \lim_{\alpha \rightarrow 0} \beta (\cancel{\alpha \mathbf{I}} + \beta \Phi^T \Phi)^{-1} \Phi^T \mathbf{t} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t} = \mathbf{w}_{ML}$

uncertain prior $\rightarrow$ ML solution

- Infinitely narrow prior ($\alpha \rightarrow \infty$):

- $\lim_{\alpha \rightarrow \infty} \mathbf{m}_N = \lim_{\alpha \rightarrow \infty} \beta (\alpha \mathbf{I} + \cancel{\beta \Phi^T \Phi})^{-1} \Phi^T \mathbf{t} = \lim_{\alpha \rightarrow \infty} \frac{\beta}{\alpha} \Phi^T \mathbf{t} = \mathbf{0} = \mathbf{m}_0$

- $\lim_{\alpha \rightarrow \infty} \mathbf{S}_N = \lim_{\alpha \rightarrow \infty} (\alpha \mathbf{I} + \cancel{\beta \Phi^T \Phi})^{-1} = \lim_{\alpha \rightarrow \infty} \frac{1}{\alpha} \mathbf{I} = \mathbf{0}$ (zero matrix)



Machine Learning 1

Lecture 4.4 - Supervised Learning
Bayesian Linear Regression - **Sequential
Bayesian Learning**

Erik Bekkers

(Bishop 3.3.1)

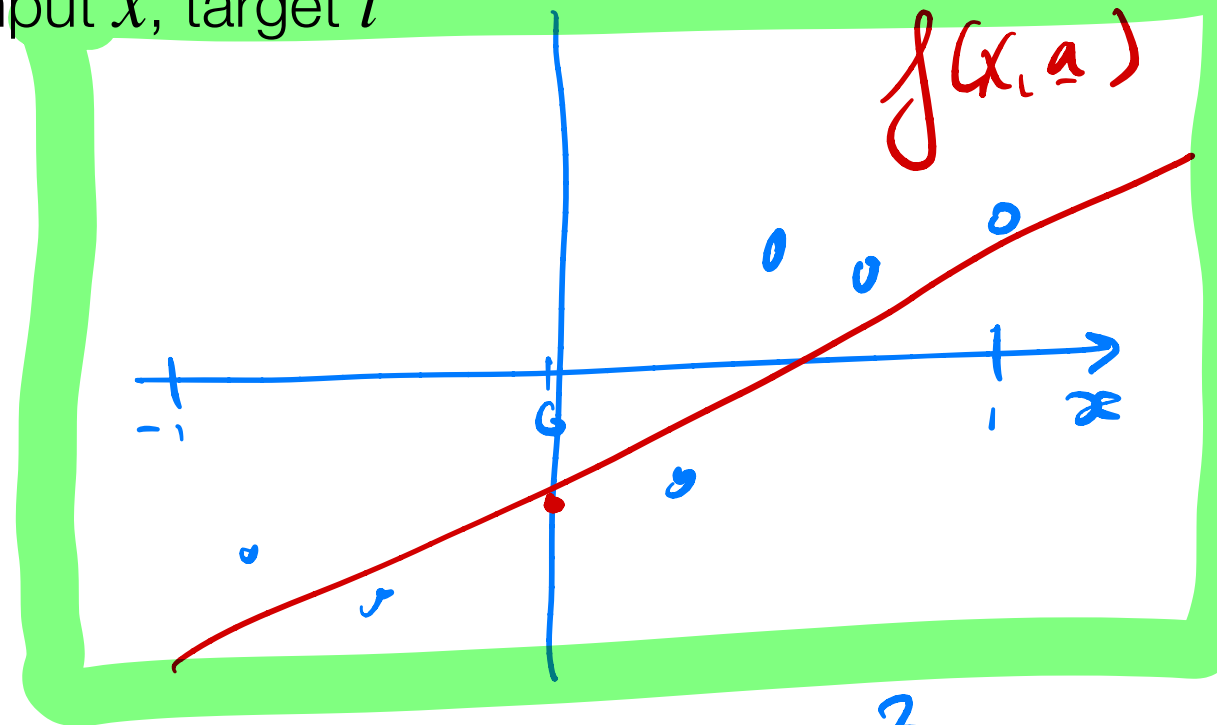


Example: Sequential Bayesian Learning

- ▶ Data come in as a sequences of observations of input x , target t

- ▶ **Synthetic data generated by**

- ▶ $x \sim \mathcal{U}(x | -1, 1)$
- ▶ $t = f(x, \mathbf{a}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.2^2)$
- ▶ $f(x, \mathbf{a}) = a_0 + a_1 x, \quad \underline{a_0 = -0.3}, \underline{a_1 = 0.5}$



- ▶ **Modeling choices**

- ▶ Target distribution: $p(t' | x', \mathbf{w}, \beta) = \mathcal{N}(t' | y(x', \mathbf{w}), \beta^{-1}), \quad \beta^{-1} = 0.2^2$
- ▶ Linear model: $y(x, \mathbf{w}) = w_0 + w_1 x$ (③ ideal w_0, w_1 ?)
- ▶ Prior: $p(\mathbf{w} | \alpha) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}), \quad \alpha = 2$

- ▶ **Sequential Bayesian Learning:** Posterior after $N - 1$ observations is prior for arrival of N^{th} datapoint! ($N=2$) $D_N = \{x_1, x_2, \dots, x_N\}$

$$p(\underline{w} | D_N) = \frac{p(x_N | \underline{w}) p(\underline{w} | D_{N-1})}{p(x_N)}$$

$N=2$:

$$p(\underline{w} | x_2, x_1) = \frac{p(x_2, x_1 | \underline{w}) p(\underline{w})}{p(x_2, x_1)}$$

prod rule

$$= \frac{p(x_2 | x_1, \underline{w}) p(x_1 | \underline{w}) p(\underline{w})}{p(x_2) p(x_1)} \quad \text{i.i.d.}$$

$$= \frac{p(x_2 | \underline{w}) p(x_1 | \underline{w}) p(\underline{w})}{p(x_2) p(x_1)}$$

posterior after x_1

$$= \frac{p(x_2 | \underline{w}) p(\underline{w} | x_1)}{p(x_2)}$$

So posterior after $N-1$ observations acts as prior for fitting with new point

Example: Sequential Bayesian Learning

- Data generated by

$$t = -0.3 + 0.5x + \epsilon$$

- Prior

$$p(\mathbf{w} | \alpha) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I})$$

- Sample 1 datapoint ★

$$(x_1, t_1)$$

- Likelihood

$$p(t_1 | x_1, \mathbf{w}, \beta) = \mathcal{N}(t_1 | w_0 + w_1 x_1, \beta^{-1})$$

- Posterior

$$p(\mathbf{w} | x_1, t_1, \alpha, \beta) \propto$$

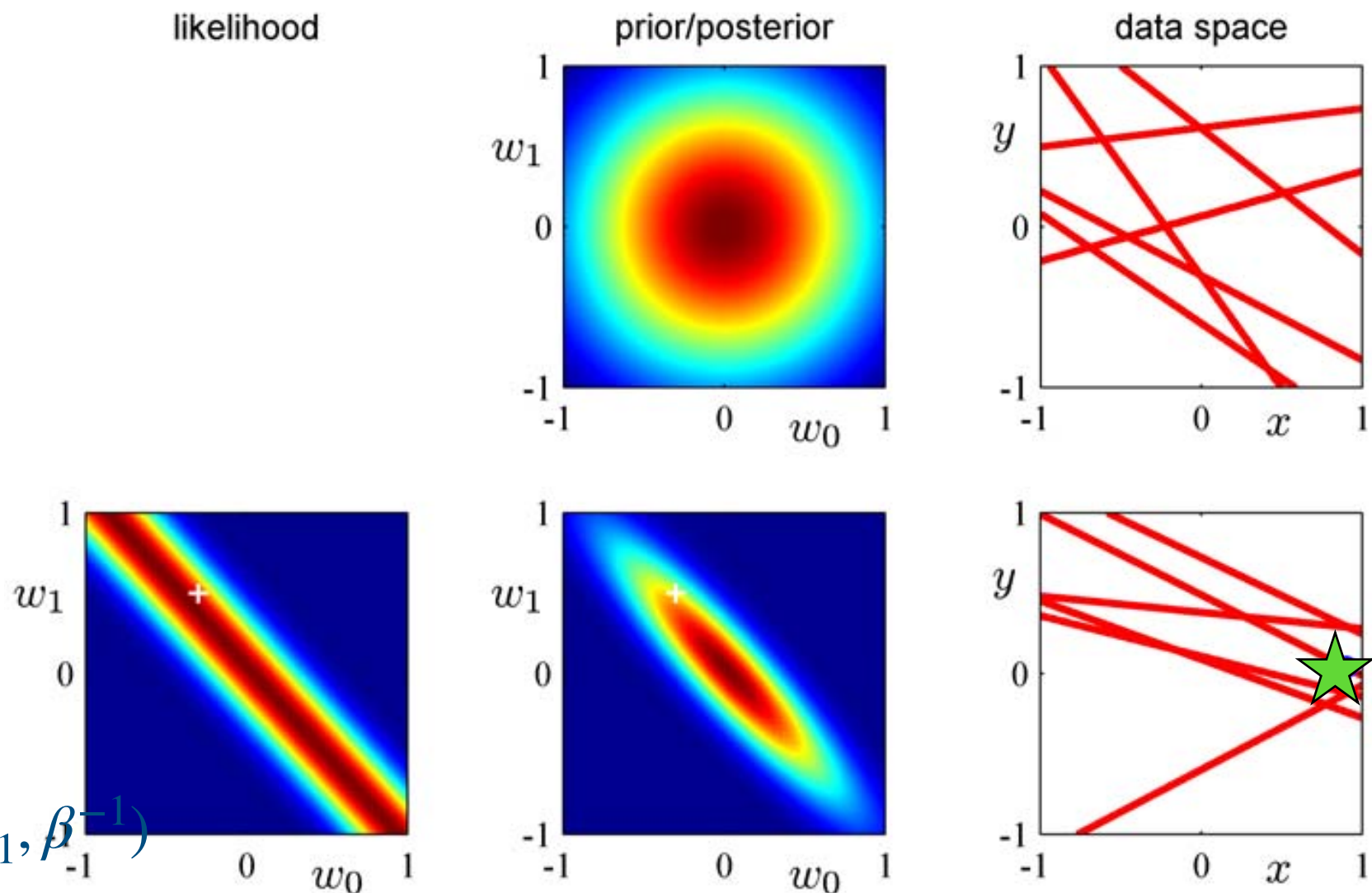


Figure: Sequential Bayesian learning (Bishop 3.7)

Sequential Bayes

<https://ideal-machine-intelligence.eu/teaching/ml1/explainers/sequential-bayes.html>

Example: Sequential Bayesian Learning

- Data generated by

$$t = -0.3 + 0.5x + \epsilon$$

- Prior

$$p(\mathbf{w} | \alpha) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I})$$

- Sample 1 datapoint ★

(x_1, t_1)

- Likelihood

$$p(t_1 | x_1, \mathbf{w}, \beta) = \mathcal{N}(t_1 | w_0 + w_1 x_1, \beta^{-1})$$

- Posterior

$$p(\mathbf{w} | x_1, t_1, \alpha, \beta) \propto p(t_1 | x_1, \mathbf{w}, \beta) \cdot p(\mathbf{w} | \alpha)$$

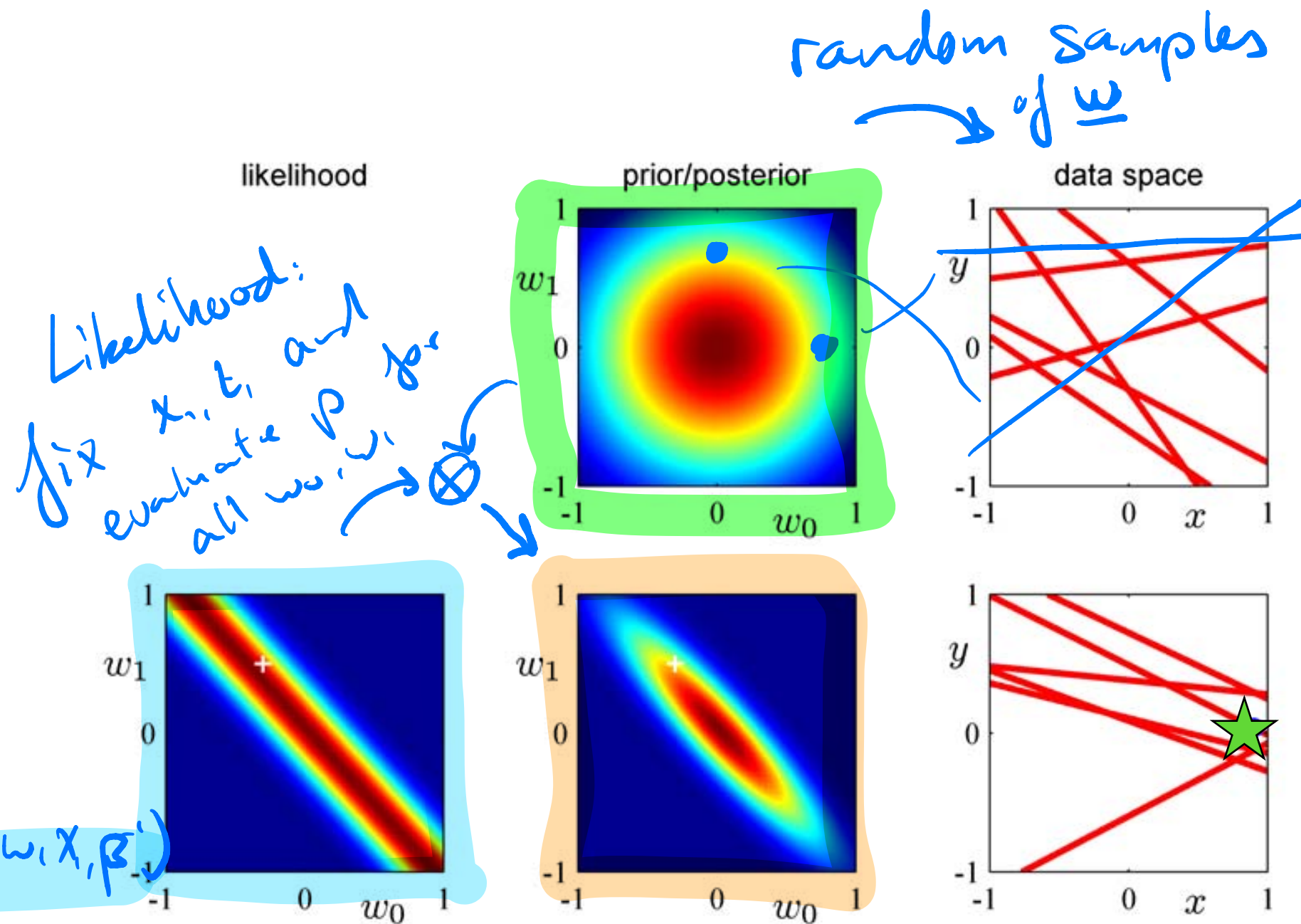


Figure: Sequential Bayesian learning (Bishop 3.7)

Example: Sequential Bayesian Learning

- Sample 2nd data point ★

(x_2, t_2)

- Posterior $\rightarrow$ Prior

- Likelihood

$$p(t_2 | x_2, \mathbf{w}, \beta)$$

- Posterior

narrows down to correct solution

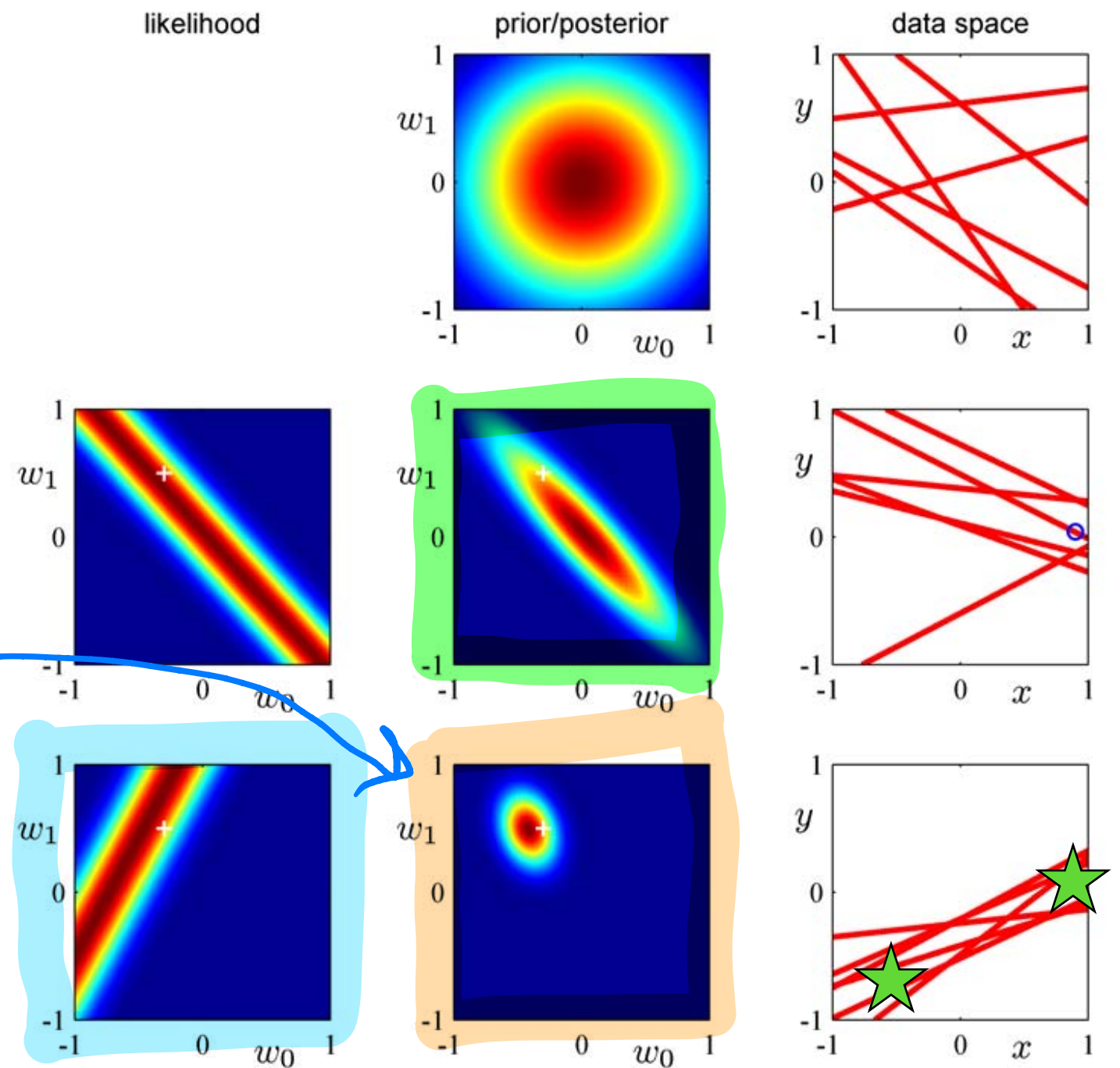


Figure: Sequential Bayesian learning (Bishop 3.7)

$$p(\mathbf{w} | (x_1, t_1), (x_2, t_2), \alpha, \beta) \propto p(t_2 | x_2, \mathbf{w}, \beta) p(\mathbf{w} | (x_1, t_1), \alpha, \beta)$$

Example: Sequential Bayesian Learning

- After 19 data points

$$D = \{(x_i, t_i)\}_{i=1}^{19}$$

- Prior

$$p(\mathbf{w} | \{(x_n, t_n)\}_{n=1}^{19}, \alpha, \beta)$$

- Likelihood

$$p(t_{20} | x_{20}, \mathbf{w}, \beta)$$

$$N \rightarrow \infty$$

$\Rightarrow$ very sharp posterior!

- Posterior

$$p(\mathbf{w} | \{(x_n, t_n)\}_{n=1}^{20}, \alpha, \beta) \propto p(t_{20} | x_{20}, \mathbf{w}, \beta) p(\mathbf{w} | \{(x_n, t_n)\}_{n=1}^{19}, \alpha, \beta)$$

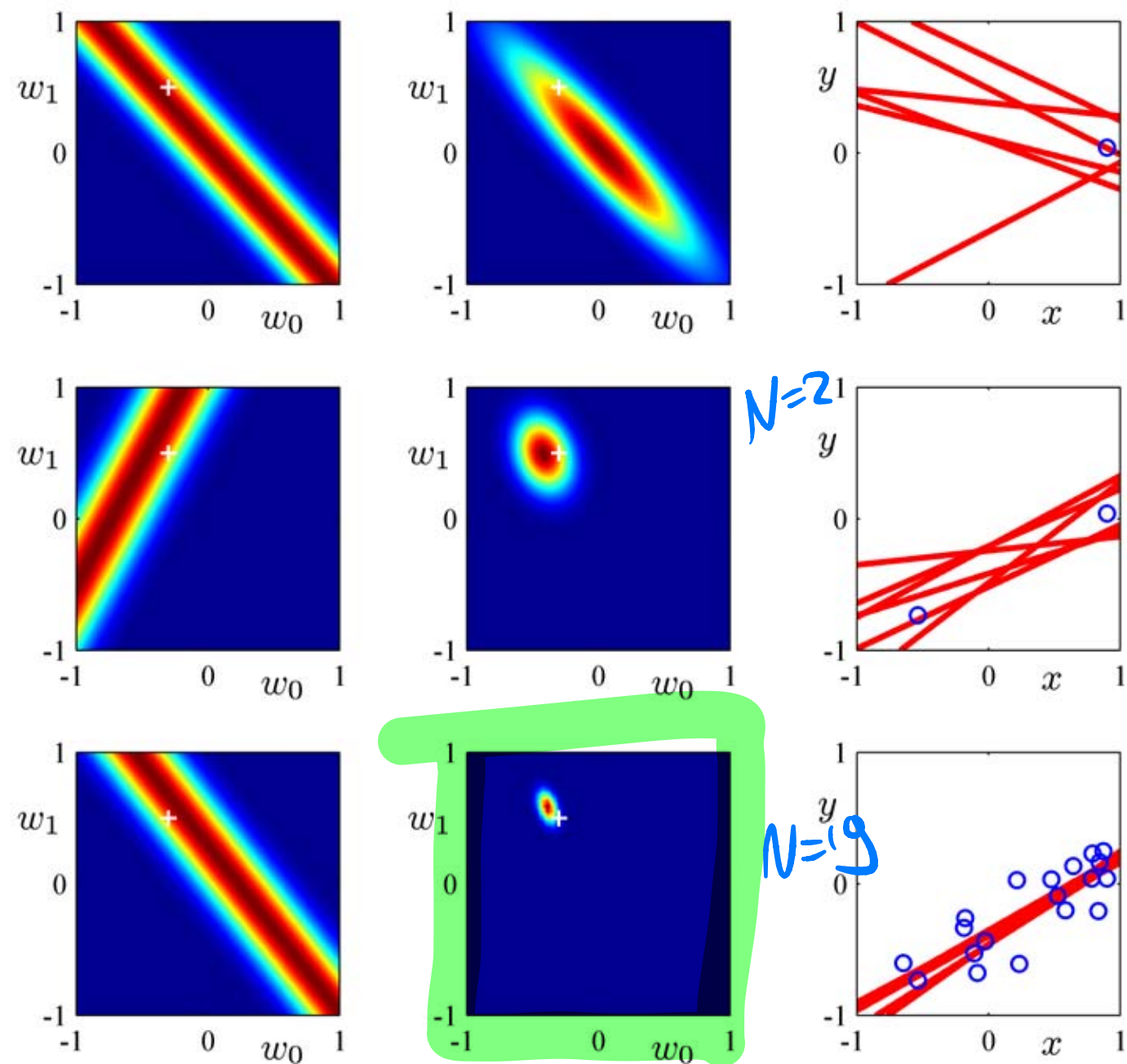


Figure: Sequential Bayesian learning (Bishop 3.7)



Machine Learning 1

Lecture 4.5 - Supervised Learning
Bayesian Linear Regression - **Predictive
Distribution**

Erik Bekkers

(Bishop 3.3.2)



Predictive Distribution

- Observed dataset with inputs $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T$ and targets $\mathbf{t} = (t_1, \dots, t_N)^T$
- Gaussian Posterior distribution (from Gaussian prior and Gaussian likelihood)

$$p(\mathbf{w} | \mathbf{X}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(\mathbf{w} | \mathbf{m}_N, \mathbf{S}_N) \quad \text{with} \quad \mathbf{m}_N = \beta \mathbf{S}_N \Phi^T \mathbf{t}$$

$$\mathbf{S}_N^{-1} = \alpha \mathbf{I} + \beta \Phi^T \Phi$$

- Parametrized Gaussian predictive distribution:

$$p(t' | \mathbf{x}', \mathbf{w}, \beta) = \mathcal{N}(t' | \phi(\mathbf{x}')^T \mathbf{w}, \beta^{-1})$$

- Gaussian Bayesian predictive distribution for new input

$$p(t' | x', \mathbf{X}, \mathbf{t}, \alpha, \beta) = \int \mathcal{N}(t' | \phi(\mathbf{x}')^T \mathbf{w}, \beta^{-1}) \mathcal{N}(\mathbf{w} | \mathbf{m}_N, \mathbf{S}_N) d\mathbf{w}$$

$$= \mathcal{N}(t' | \mathbf{x}', \phi(\mathbf{x}')^T \mathbf{m}_N, \sigma_N^2(\mathbf{x}'))$$

Bishop Eq.
2.115

- With $\sigma_N^2(\mathbf{x}') = \beta^{-1} + \phi(\mathbf{x}')^T \mathbf{S}_N \phi(\mathbf{x}')$

analytic position dependent
variance

MAP solution

model averaging
with posterior

integral w
Gaussians gives
again a Gaussian

Predictive Distribution

active learning?

$N=1$

$N=2$

▶ Datasets:

- ▶ $t = \sin(2\pi x) + \epsilon$
- ▶ $\epsilon \sim \mathcal{N}(0, \beta^{-1})$

▶ Dataset sizes:

- ▶ $N = 1, 2, 4, 25$

▶ Model:

- ▶ $y(x, \mathbf{w}) = \boldsymbol{\phi}(x)^T \mathbf{w}$
- ▶ $\phi_j(x)$: Gaussian basis functions

mean $y(x, \mathbf{w})$
given by MAP

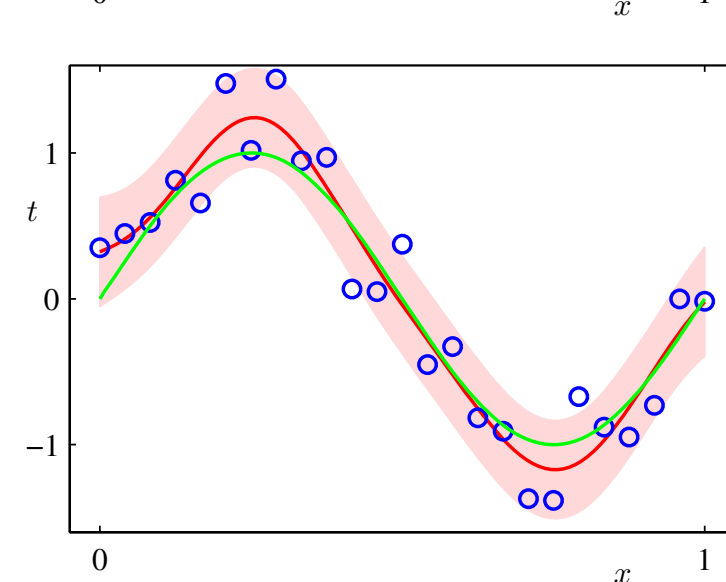
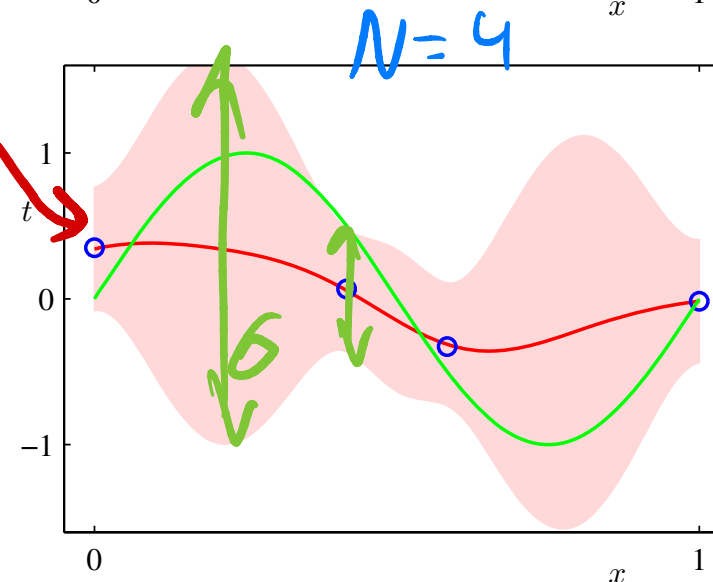
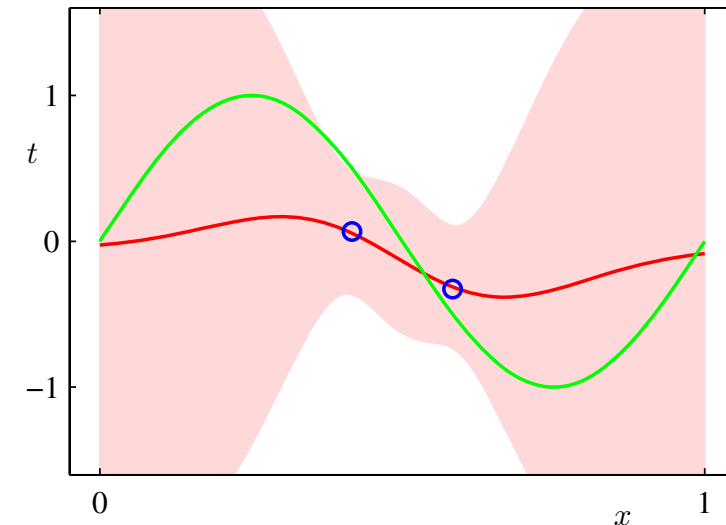
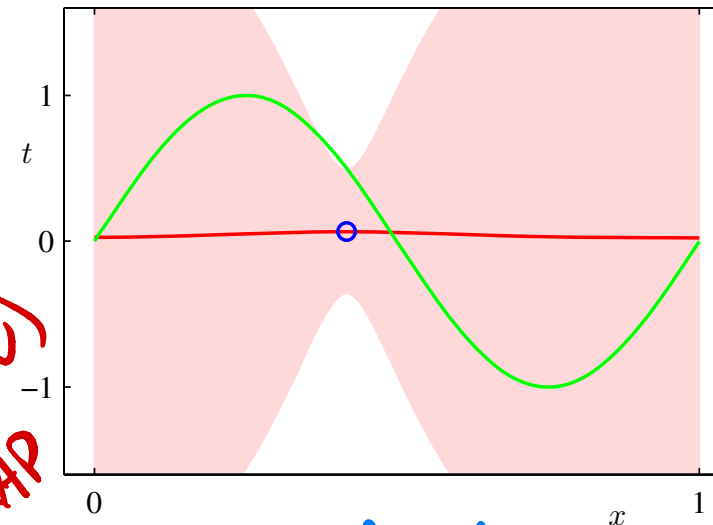


Figure: Predictive distribution (Bishop 3.8)

▶ Predictive distribution:

$$p(t' | x', \mathbf{X}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(t' | \underbrace{y(x', \mathbf{w}_{\text{MAP}})}_{\boldsymbol{\phi}(\mathbf{x}')^T \mathbf{m}_N}, \sigma_N^2(\mathbf{x}'))$$

$$\sigma_N^2(\mathbf{x}') = \beta^{-1} + \boldsymbol{\phi}(\mathbf{x}')^T \mathbf{S}_N \boldsymbol{\phi}(\mathbf{x}'), \quad \mathbf{m}_N = \beta \mathbf{S}_N \boldsymbol{\Phi}^T \mathbf{t}, \quad \mathbf{S}_N^{-1} = \alpha \mathbf{I} + \beta \boldsymbol{\Phi}^T \boldsymbol{\Phi}$$

Samples drawn from Bayesian Predictive Distribution

$$\underline{\mathbf{w}} \sim p(\mathbf{w} | \mathbf{X}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(\mathbf{w} | \mathbf{m}_N, \mathbf{S}_N),$$

$$\text{with } \mathbf{m}_N = \beta \mathbf{S}_N \Phi^T \mathbf{t}$$

$$\mathbf{S}_N^{-1} = \alpha \mathbf{I} + \beta \Phi^T \Phi$$

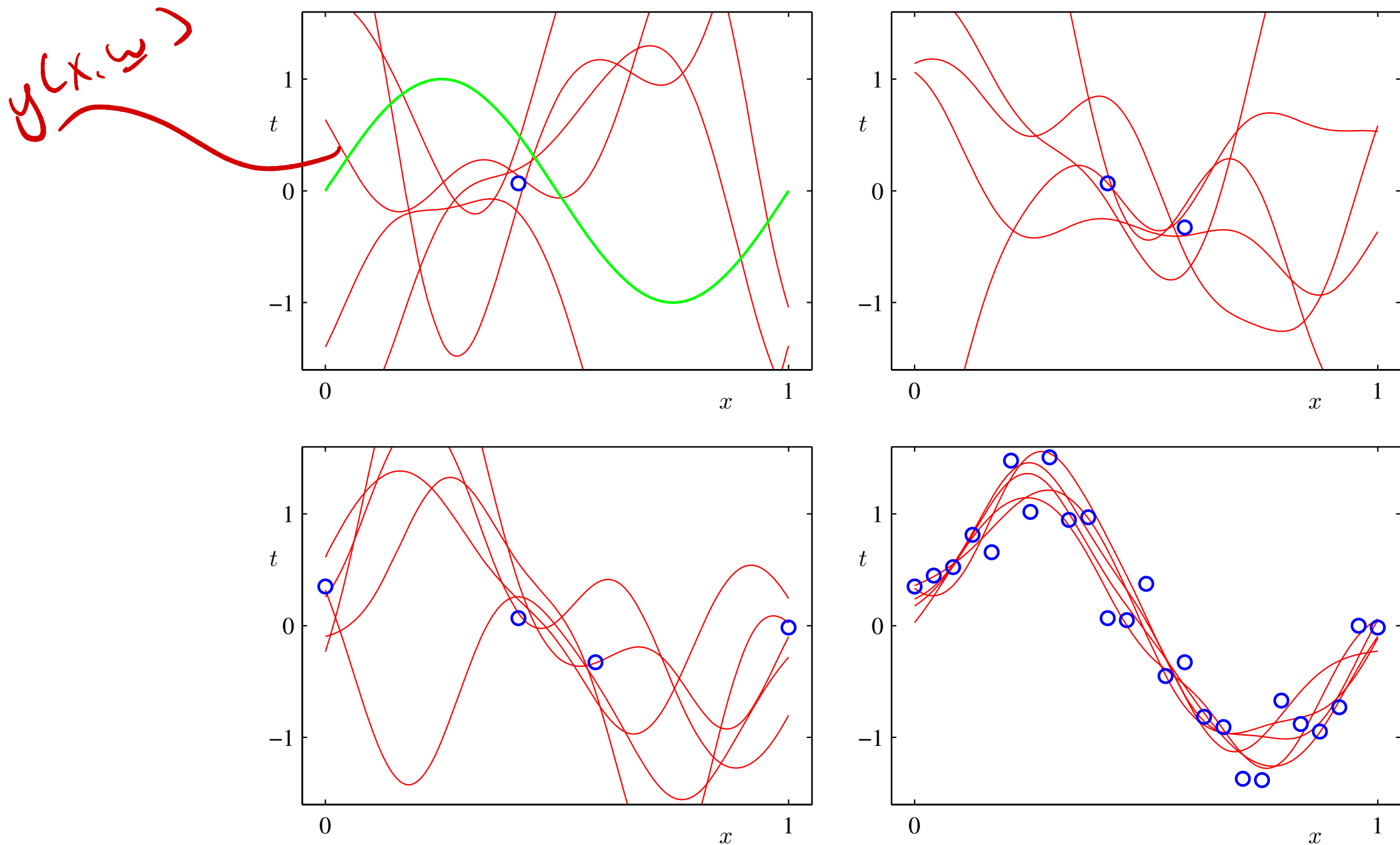


Figure: Sample functions $y(x, \mathbf{w})$ with $\mathbf{w}$ sampled from posterior distribution (Bishop 3.9)



Machine Learning 1

Lecture 3.5 - Supervised Learning

Regularized Least Squares

Erik Bekkers

(Bishop 3.1.4)



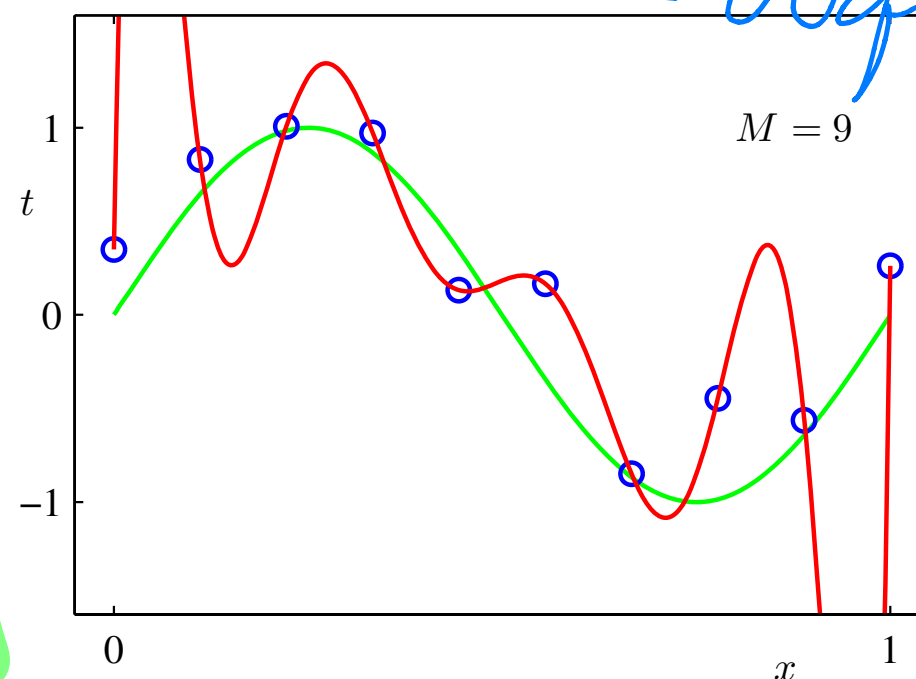
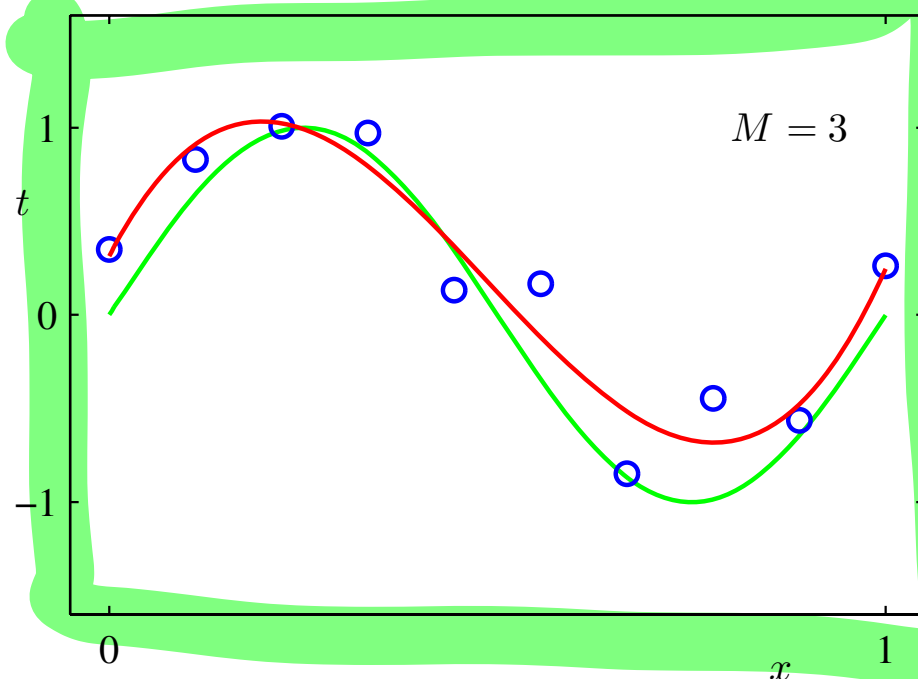
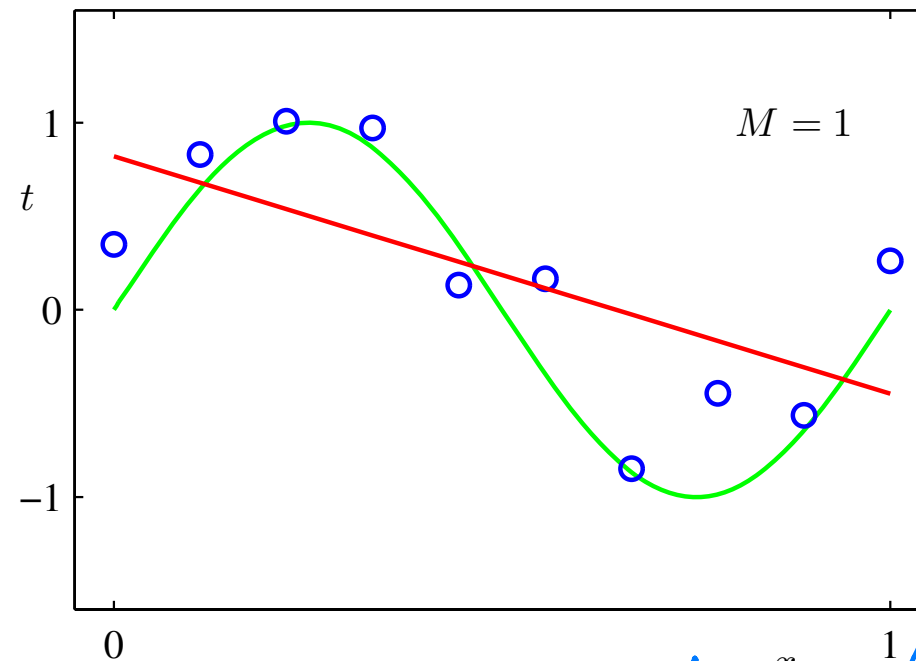
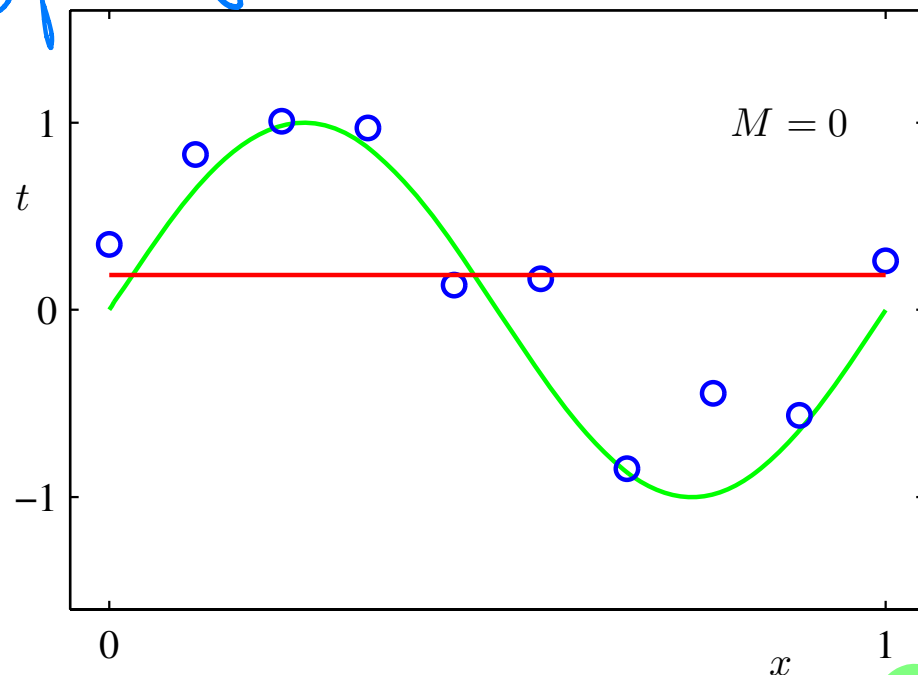
Example: Overfitting and Underfitting

$$t = \sin(2\pi x) + \varepsilon$$

$$\varepsilon \sim \mathcal{N}(0, \beta^{-1})$$

$$y(x, \mathbf{w}) = w_0 + \sum_{i=1}^M w_i x^i$$

underfitting



overfitting

Figure: Fits of different polynomials (Bishop 1.4)

Example: Overfitting (M=9)

	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

why not zero?

Table: Polynomial coefficients (Bishop 1.1)

Regularized Least Squares

$$\underline{w} = \begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_{M-1} \end{pmatrix} \in \mathbb{R}^M$$

- ▶ Instead of manually constraining the number of parameters for small datasets, **add penalty term** for large parameter values:

$$\tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^N \{t_i - y(\mathbf{x}_i, \mathbf{w})\}^2 + \frac{\lambda}{2} \sum_{i=1}^{M-1} w_i^2$$

a.k.a. - Ridge regression

- ℓ_2 regularization ($\frac{\lambda}{2} \|\underline{w}\|_2^2$)

- weight decay

- ▶ The bias term w_0 is often not included in regularization

- does not add to model complexity

- constant offset is often necessary

Parameter estimates with Gaussians

- Given **Likelihood**/Data model:

$$p(t | x, \mathbf{w}, \beta) = \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1})$$

- The **ML parameter estimate** is obtained via **least squares**:

$$\mathbf{w}_{ML} = \operatorname{argmin}_{\mathbf{w}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2$$

- Additionally, given Gaussian weight **Prior**:

$$p(\mathbf{w} | \alpha) = \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha^{-1})$$

Ridge reg. $\lambda = \frac{\alpha}{\beta}$

- The **MAP parameter estimate** is obtained via **regularized least squares**:

$$\mathbf{w}_{MAP} = \operatorname{argmin}_{\mathbf{w}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

Example: Regularized Polynomial Regression

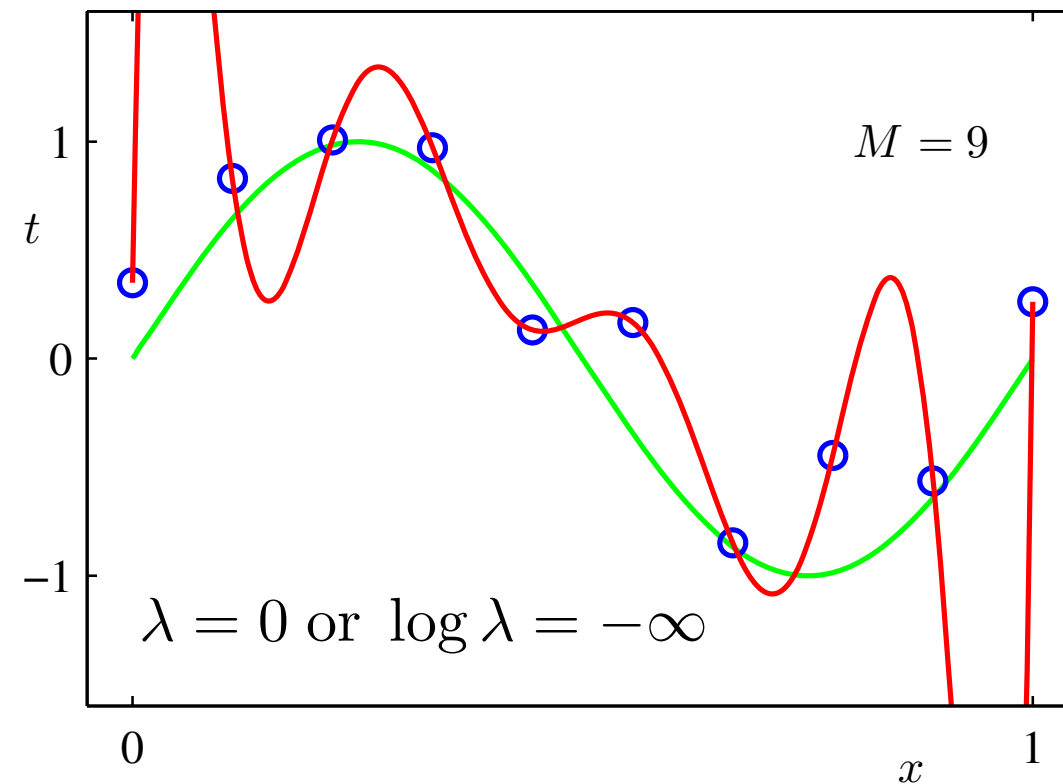


Figure: polynomial regression (Bishop 1.4)

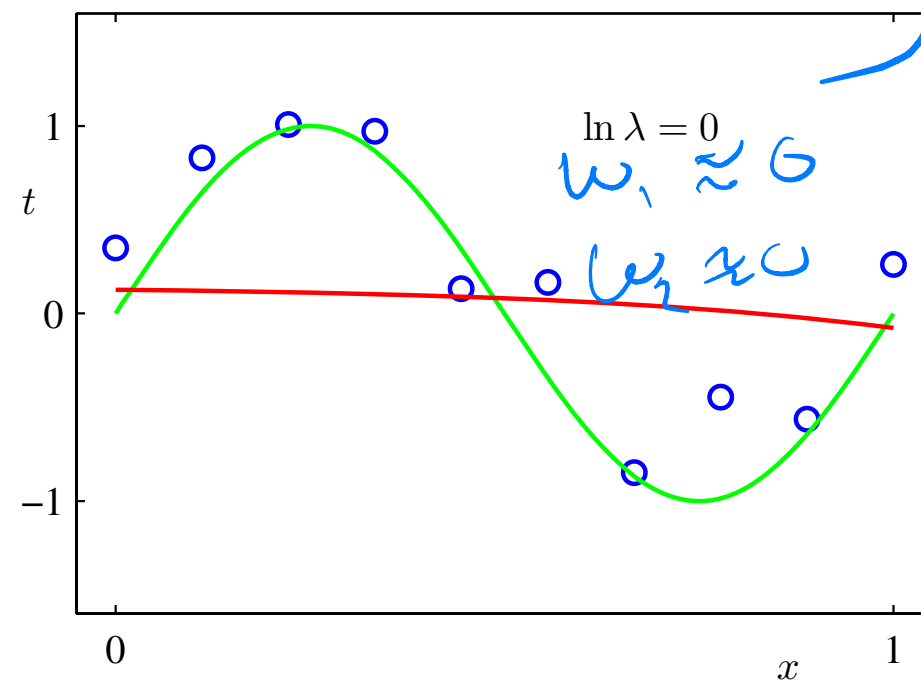
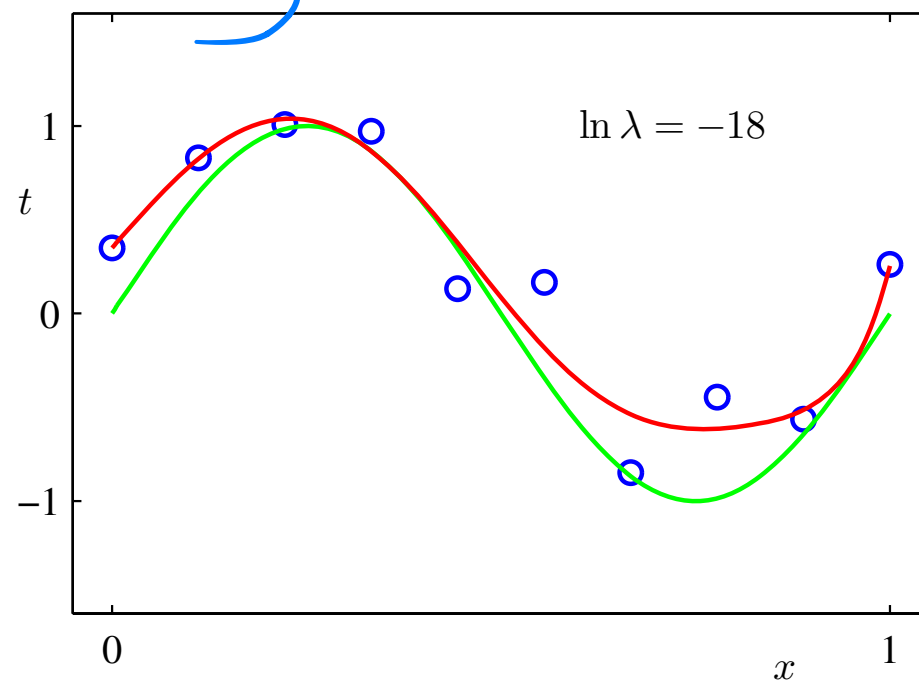


Figure: Regularized polynomial regression (Bishop 1.7)

Example: Regularized Polynomial Regression

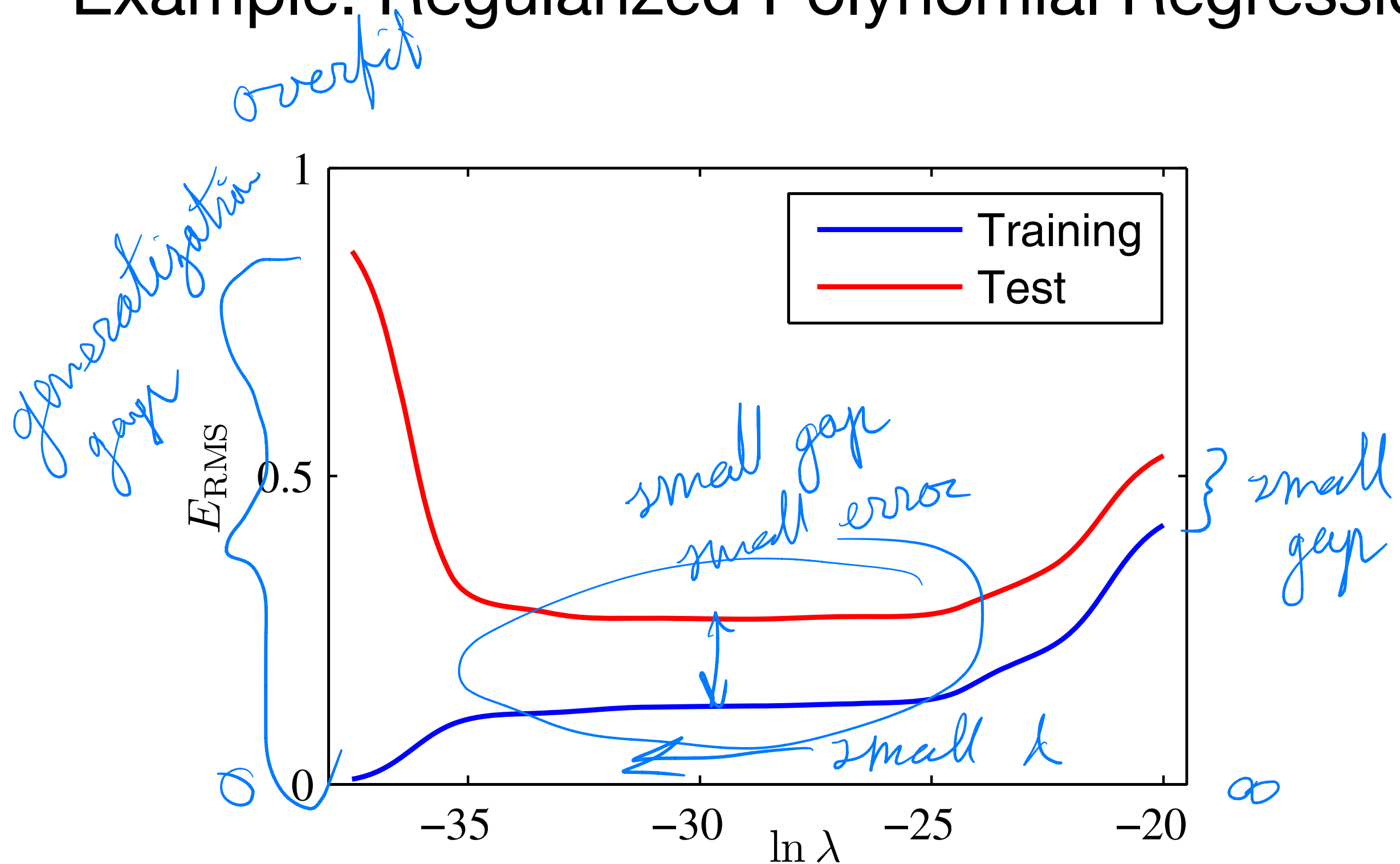


Figure: train and test errors for regularized $M=9$ polynomial regression (Bishop 1.8)

Regularized Least Squares (II)

Weight decay: $\tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\}^2 + \frac{\lambda}{2} \sum_{i=1}^{M-1} |w_i|^2$

General penalty:

$$\tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\}^2 + \frac{\lambda}{2} \sum_{i=1}^{M-1} |w_i|^q$$

Case $q = 1$: Lasso

→ Sparsification effect
→ many $w_i = 0$

Equivalent to some constraint optimization problem (Bishop App. E)

minimize $\frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\}^2$ subject to constraint $\sum_{i=1}^M |w_i|^q \leq \eta(\lambda)$ (w6)

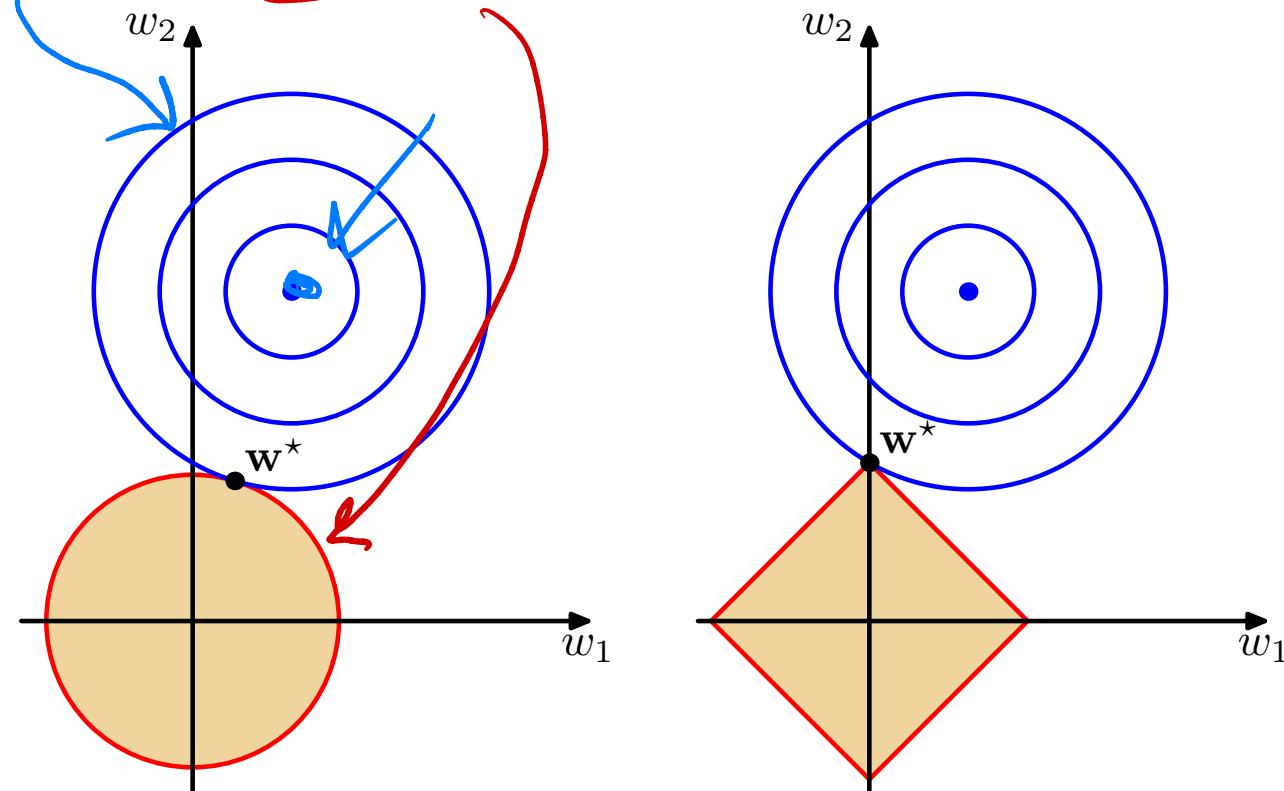


Figure: regularization as constrained optimization (Bishop 3.4)

Lagrange multiplier

Regularized Least Squares: sparse weights

$$\text{minimize } \frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\}^2 \quad \text{subject to constraint } \sum_{i=1}^M |w_i|^q \leq \eta$$

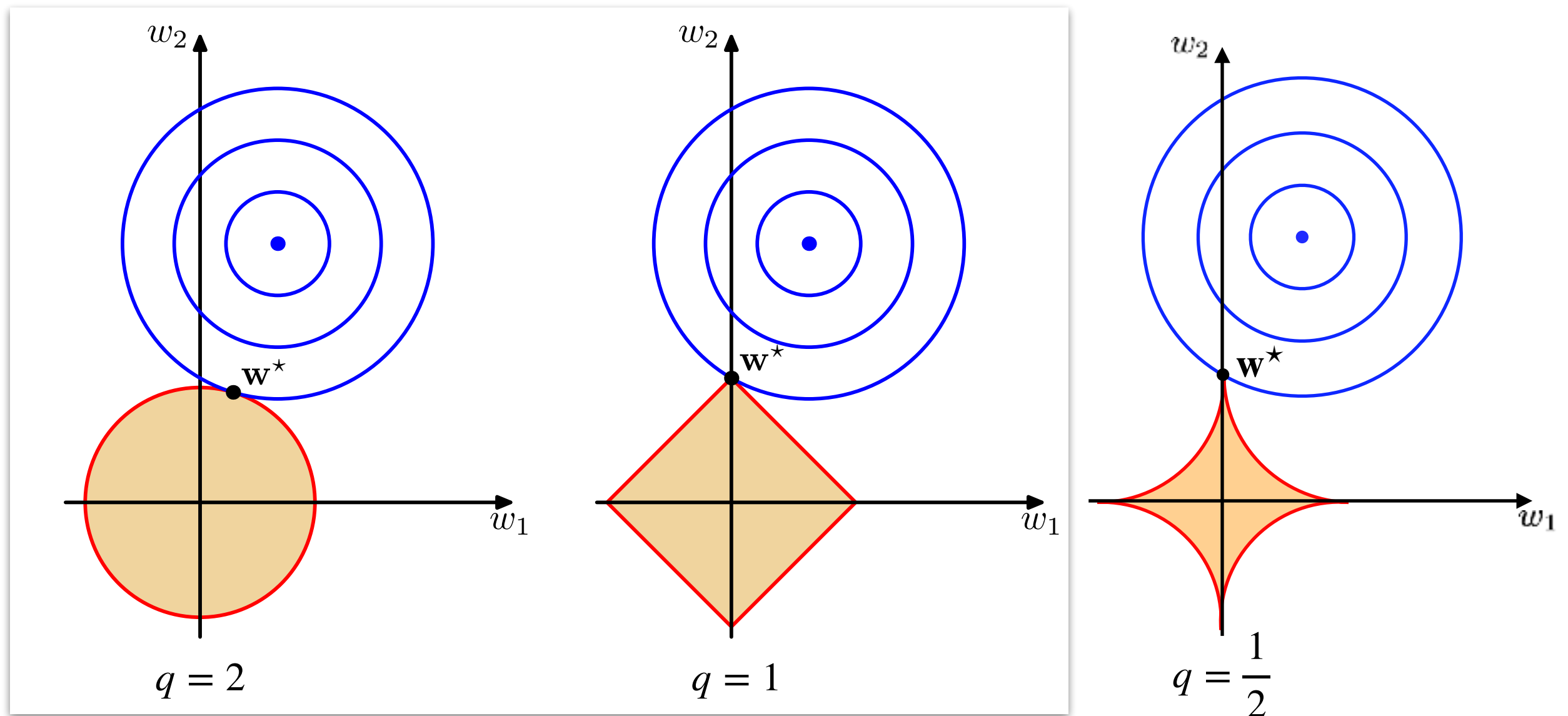
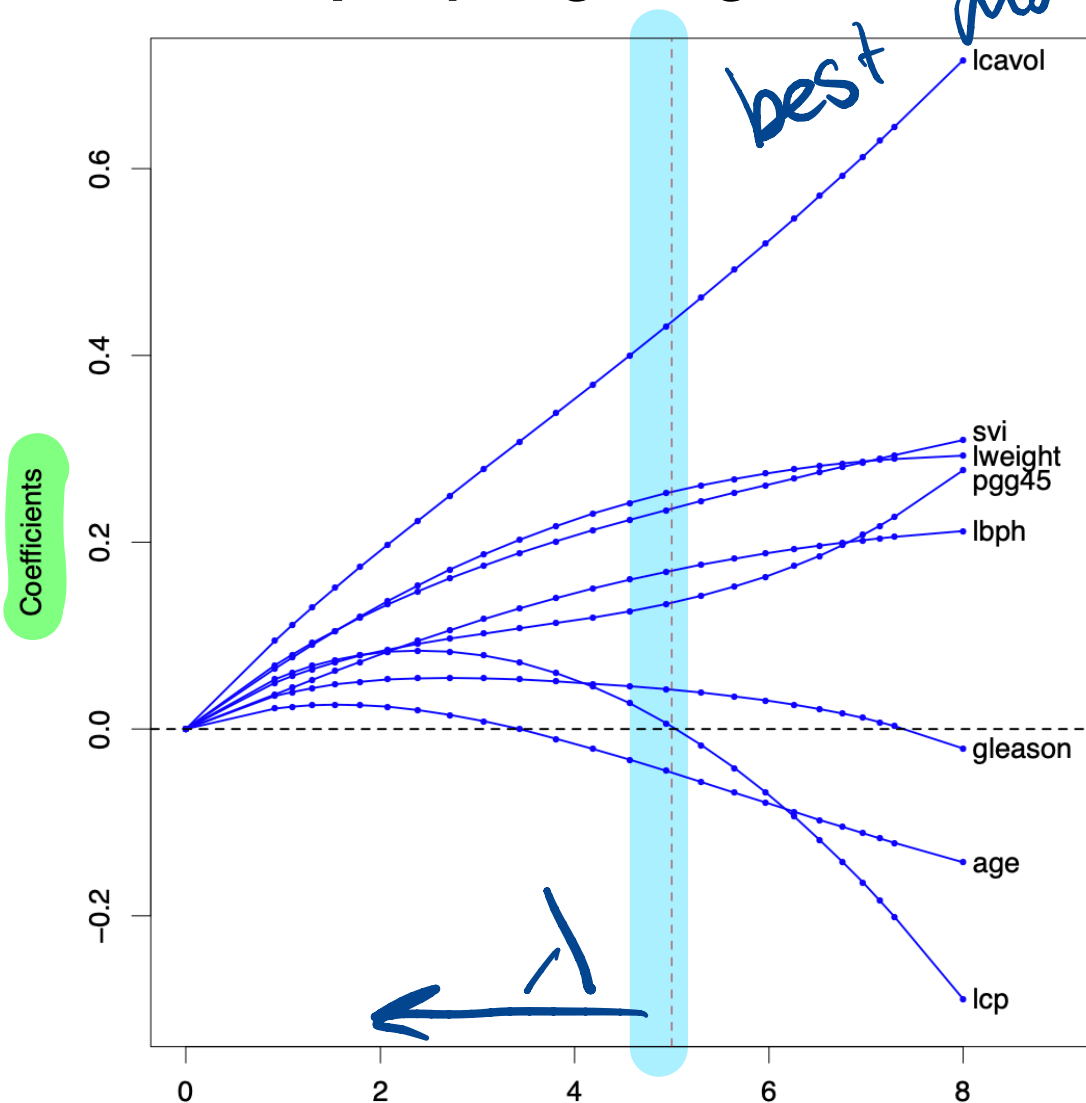


Figure: regularization as constrained optimization (Bishop 3.4)

Example: Prostate specific antigen prediction

q=2 (Ridge regression)



q=1 (Lasso regression)

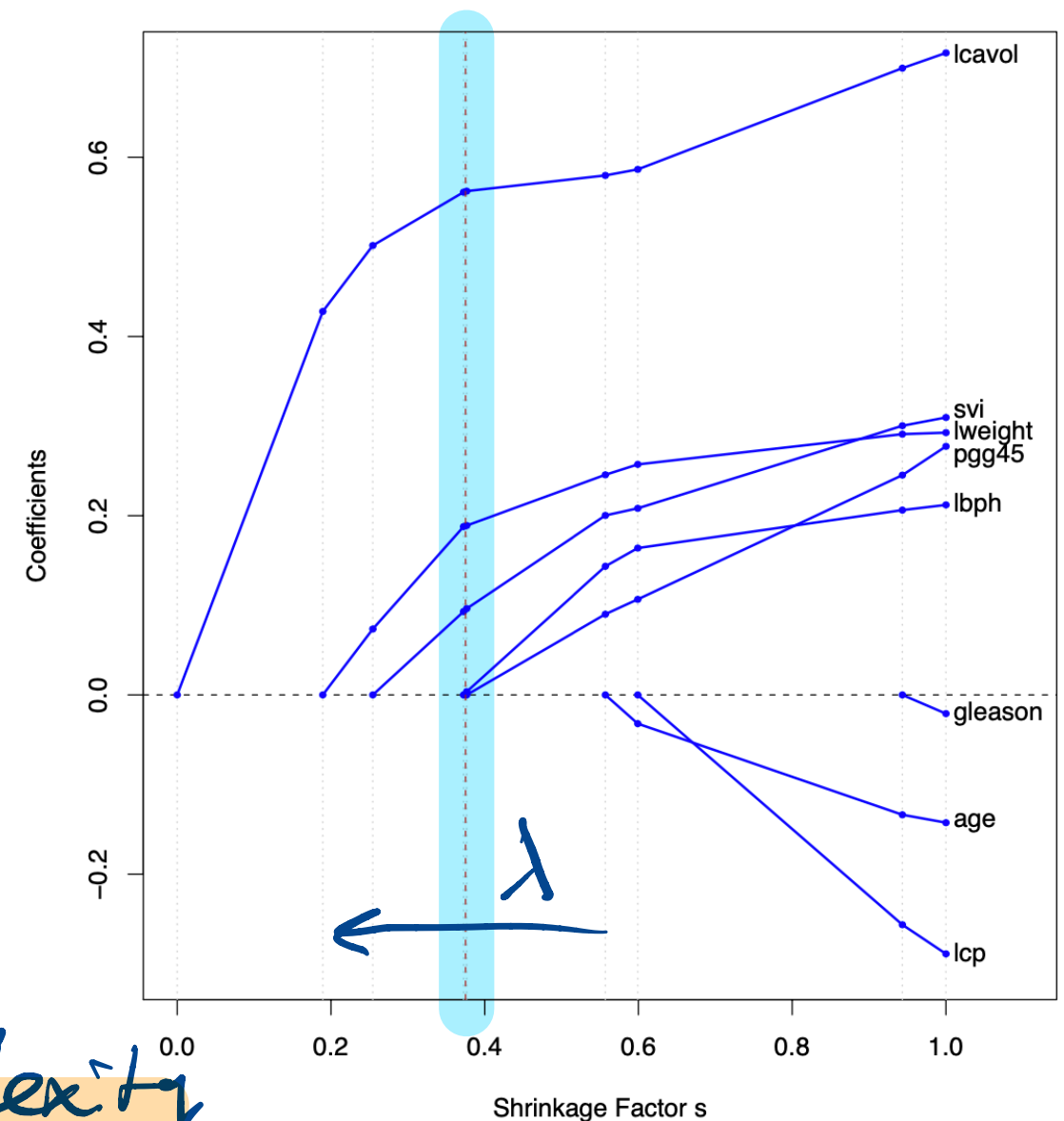


FIGURE 3.8. Profiles of ridge coefficients for the prostate cancer example, as the tuning parameter λ is varied. Coefficients are plotted versus $df(\lambda)$, the effective degrees of freedom. A vertical line is drawn at $df = 5.0$, the value chosen by cross-validation.

FIGURE 3.10. Profiles of lasso coefficients, as the tuning parameter t is varied. Coefficients are plotted versus $s = t / \sum_1^p |\hat{\beta}_j|$. A vertical line is drawn at $s = 0.36$, the value chosen by cross-validation. Compare Figure 3.8 on page 65; the lasso profiles hit zero, while those for ridge do not. The profiles are piece-wise linear, and so are computed only at the points displayed; see Section 3.4.4 for details.

Figures from the Elements of Statistical Learning (ESL - Hastie et al.)



Machine Learning 1

Lecture 4.1 - Supervised Learning

Model Selection

Erik Bekkers

(Bishop 1.3)



Supervised Learning: Evaluating Errors

- Q: How can we reliably estimate the model performance properly for unknown data?
- Q: How can we choose the optimal hyperparameters?

Model selection | Dataset splits

Divide data $D = \{(\mathbf{x}_1, t_1), \dots, (\mathbf{x}_N, t_N)\}$ in 3 groups: ①

Training set D_{train} (± 80 % of D):

- Minimize the error $E(y(\mathbf{x}, \mathbf{w}), t)$ for every $(\mathbf{x}, t) \in D_{\text{train}}$

obtain $\mathbf{w}^*$

with new different hyperparameters (1)

given hyperpar.

repeat

Validation set D_{val} (± 10 % of D):

- Used to estimate generalization error for every $(\mathbf{x}_{\text{val}}, t_{\text{val}}) \in D_{\text{val}}$

②

$E(y(\mathbf{x}_{\text{val}}, \mathbf{w}^*), t_{\text{val}})$

estimate generalization performance

Test set D_{test} (± 10 % of D):

- final test/generalization error estimate $E(y(\mathbf{x}_{\text{test}}, \mathbf{w}^*), t_{\text{test}})$ for every $(\mathbf{x}_{\text{test}}, t_{\text{test}}) \in D_{\text{test}}$

Can never be used for model selection!!!

Supervised Learning: Small Datasets

- ▶ Small dataset → small validation and test set
- ▶ Noisy model selection and estimate of generalization error
- ▶ **Cross-validation:**
 - ▶ Split data $D = \{(\mathbf{x}_1, t_1), \dots, (\mathbf{x}_N, t_N)\}$ in to K folds
 - ▶ Train model y on $K - 1$ folds (fold k left out) → $\hat{y}^{-k}$

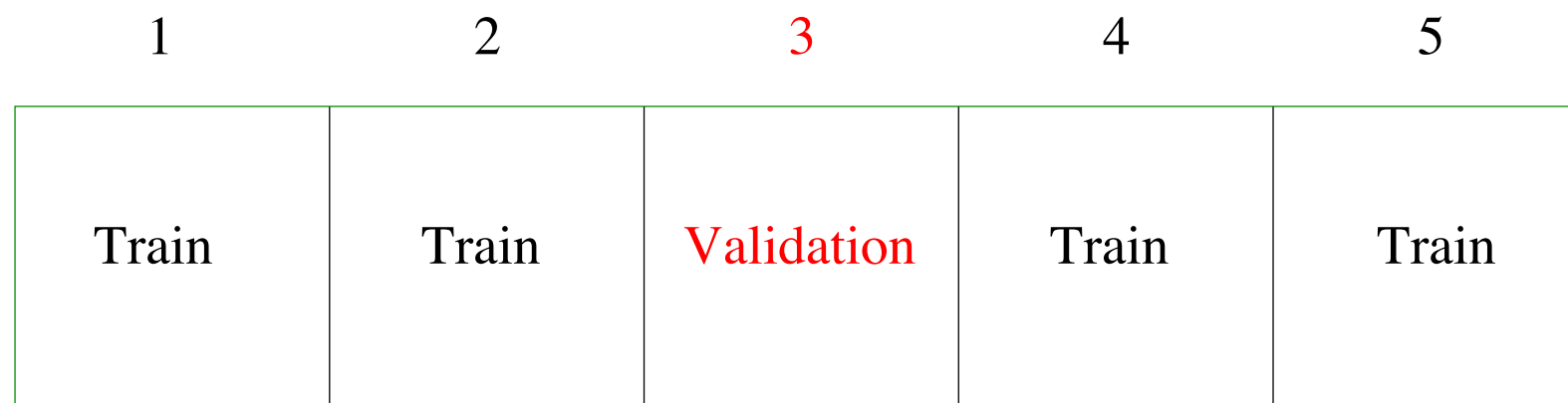


Figure: K-fold splitting of dataset (ESL 7.10)

- ▶ Leave-one-out cross validation: $K = N$

Cross-Validation

① trained on data w k^{th} split left out

② use to tell which split each of N data points is assigned to

► K trained functions $\hat{y}^{-k}$

► Use indexing function $\kappa : \{1, \dots, N\} \rightarrow \{1, \dots, K\}$

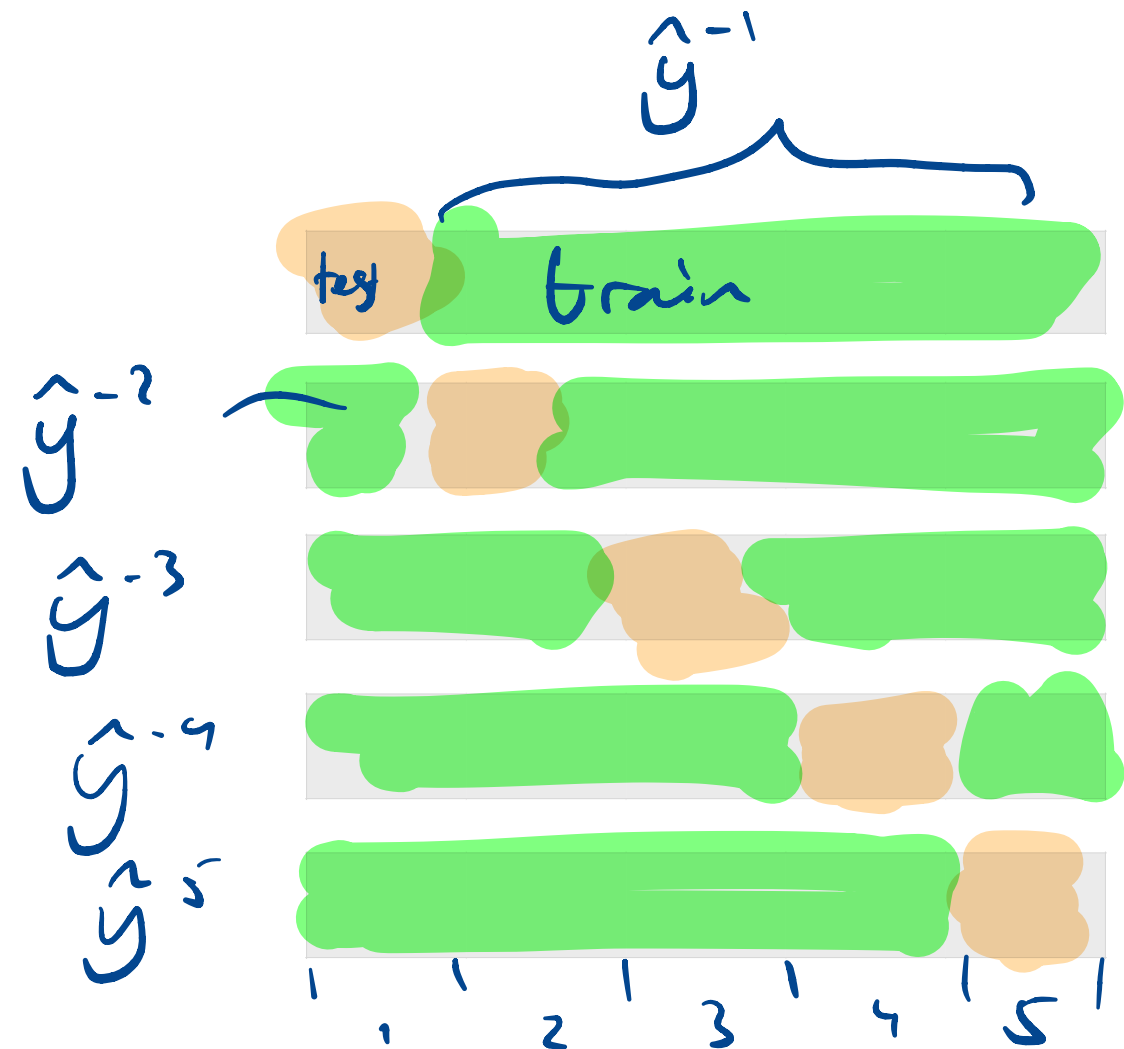
► Cross validation error:

$$CV(\hat{y}) = \frac{1}{N} \sum_{i=1}^N E(\hat{y}^{-\kappa(i)}(\mathbf{x}_i), t)$$

1. model selection

2. estimate generalization perf.

for each x_i test with model that didn't use x_i



Cross-Validation: Model Selection

- Hyperparameter selection: α

$$CV(\hat{y}_\alpha) = \frac{1}{N} \sum_{i=1}^N E(\hat{y}_\alpha^{-k(i)}(\mathbf{x}_i), t)$$

$$\alpha^* = \underset{\alpha}{\operatorname{argmax}} CV(\hat{y}_\alpha)$$

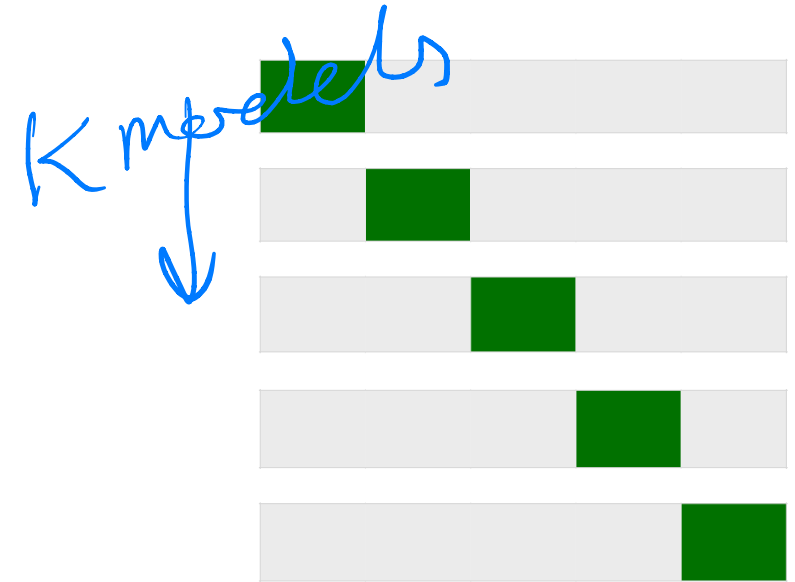
- Multiple hyperparameters: $\alpha \in \{\alpha_1, \alpha_2\}, \beta \in \{\beta_1, \beta_2, \beta_3\}$

- How many times should CV be performed?

$$2 \times 3 = 6$$

- Total number of training runs?

$$6 \times K$$

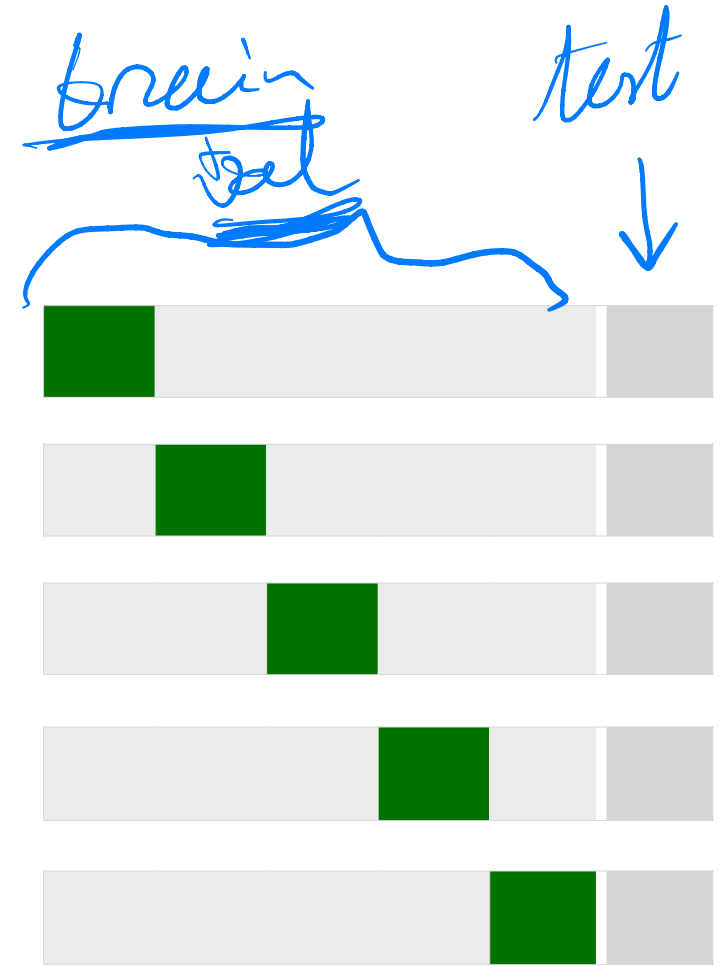


Cross-Validation: Test Error Estimation

- ▶ After Model selection we obtain some

$$\alpha^*, \beta^*$$

- ▶ Retrain y on all train data with α^*, β^*
- ▶ Evaluate model on held-out test set
 - ▶ Still noisy estimate of generalization error if test set is small
- ▶ Nested cross validation!



Nested Cross-Validation

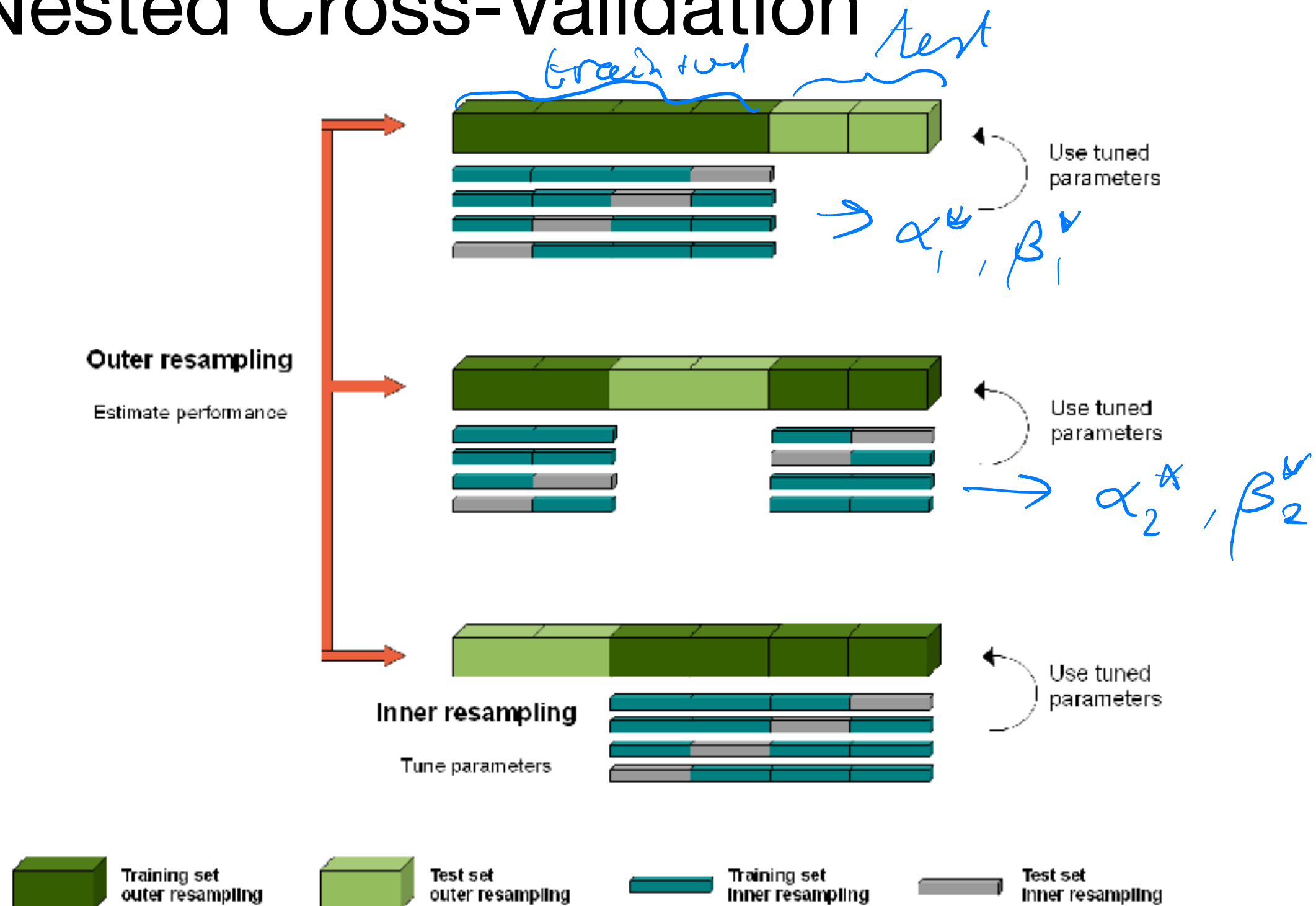


Figure: Nested cross-validation

https://mlr-org.github.io/mlr-tutorial/devel/html/nested_resampling/index.html
(site is offline unfortunately)



Machine Learning 1

Lecture 4.2 - Supervised Learning

Bias Variance Decomposition

Erik Bekkers

(Bishop 1.5.5, 3.2)





Machine Learning 1

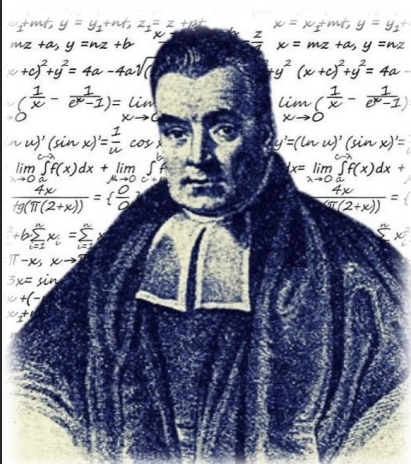
Lecture 4.2 - Supervised Learning

Bias Variance Decomposition

Erik Bekkers

(Bishop 1.5.5, 3.2)

The probabilistic view



W7

Gaussian Processes

W6

Kernel Methods

W5

Latent variable models

W4

Neural Networks

W3

Classification

W2

Regression

W1

Probabilistic modeling

Random variables, distributions,
maximum likelihood estimation,
MAP

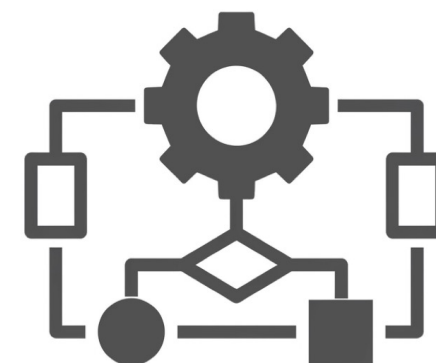
Optimization

Exact/analytic, (stochastic
gradient descent, constrained
optimization, 2nd order)

Mathematical tools

Multi-variate calculus, linear
algebra

The algorithmic view



Expected Loss for Regression $\mathbb{E}_{(\mathbf{x},t) \sim p(\mathbf{x},t)}[L(t, y(\mathbf{x}))]$

Why do models make errors?

What kind of errors can we expect?

- Consider dataset of observations $(\mathbf{x}, t) \sim p(\mathbf{x}, t)$ and a model $y(\mathbf{x})$

$$D = \{(x_1, t_1), (x_2, t_2), \dots\}, 5$$

- The model makes errors (regression loss function):

$$L(t, y(\mathbf{x})) = (t - \hat{t})^2$$

- Every time we make an observation of random variables $(\mathbf{x}, t)$ we make a different error. We now consider the expected loss:

$$\mathbb{E}_{(\mathbf{x},t) \sim p(\mathbf{x},t)}[L(t, y(\mathbf{x}))] = \iint (t - y(\mathbf{x}))^2 p(\mathbf{x}, t) d\mathbf{x} dt$$

- The best model y we can possibly have (Bishop 1.5.5):

$$y(\mathbf{x}) = \mathbb{E}[t | \mathbf{x}]$$

Regression function

Example: $t = \sin 2\pi x + \epsilon$, with $\epsilon \sim \mathcal{N}(0, \beta^{-1})$

